

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

3.0 TDV6 ENGINE CONTROL MODULE
(ECM) - P0700-00 TO P22FE-85

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable.
Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the warranty policy and procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system)
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places and with a current calibration certificate. When testing resistance, always take the resistance of the digital multimeter leads into account
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- If diagnostic trouble codes are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals
- Where an 'on demand self-test' is referred to, this can be accessed via the 'diagnostic trouble code monitor' tab on the manufacturers approved diagnostic system
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

The table below lists all diagnostic trouble codes (DTCs) that could be logged in the rear integrated control panel, for additional diagnosis and testing information refer to the relevant diagnosis and testing section. For additional information, refer to: [Electronic Engine Controls](#) (303-14A Electronic Engine Controls - TDV6 3.0L Diesel, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0700-00	Transmission Control System (MIL Request) - No sub type information	MIL request by automatic gearbox	NOTE: If PCM has just been updated or newly programmed, then the ignition will need to be cycled (switched on and off) 3 times. Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control module for related DTCs and refer to the relevant DTC index
P0703-62	Brake Switch "B" Circuit - Signal compare failure	- The engine control module detected failure when comparing two or more input parameters for plausibility - The brake pedal switch signal received over CAN is defective - Brake pedal switch detached from pedal box whilst electrically connected - Brake pedal switch adjustment incorrect - Brake pedal switch harness failure - Connector is disconnected, connector pin is	- Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test - Check engine control module for related DTCs and refer to relevant DTC index - Refer to the electrical circuit diagram and check the brake pedal switch circuits for open circuit, short circuit to ground, short circuit to power - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new brake pedal switch as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest

		backed out, connector pin corrosion ▪ Brake pedal switch failure	
P0850-00	Park/Neutral Switch Input Circuit - No sub type information	▪ Park neutral switch circuit between transmission control switch, central junction box and engine control module short circuit to ground, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Refer to the electrical circuit diagram and check the park neutral switch circuit between transmission control switch, central junction box and engine control module for short circuit to ground, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P0850-64	Park/Neutral Switch Input Circuit - Signal plausibility failure	▪ The engine control module detected plausibility failures ▪ Park neutral switch circuit between transmission control switch, central junction box and engine control module short circuit to ground, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Refer to the electrical circuit diagram and check the park neutral switch circuit between transmission control switch, central junction box and engine control module for short circuit to ground, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P0A08-00	DC/DC Converter Status Circuit - No sub type information		▪ Check engine control module for related DTCs and refer to relevant index ▪ Using the Jaguar Land Rover approved diagnostic equipment reprogram the

		NOTE: To monitor initialisation of fuel injector supply DC/DC converter bank 1 and 2. Error detection of DC/DC buffer capacitor during start up of system - Other related DTCs - Fuel injector coding incorrect - Harness failure - Wiring integrity - Fuel injector circuit short circuit to power - Vehicle battery over charged - Charging system fault - over charging - Connector is disconnected, connector pin is backed out, connector pin corrosion - Engine control module power and ground circuits open circuit, high resistance - Engine control module mechanical integrity	injector codes - Refer to the electrical circuit diagram and check connections are secure and wiring integrity - Refer to the electrical circuit diagram and check the fuel injector circuit for short circuit to power - Check the vehicle charging system performance to ensure the voltage regulation is correct - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Check engine control module for mechanical integrity - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tool and retest
P0A0F-00	Engine Failed to Start - No sub type information	The engine control module has detected that the operation of a component is	- Check engine control module for related DTCs and refer to relevant index - Refer to the electrical circuit diagram

		prevented by an obstruction - Engine control module has logged an extended crank with no engine start - Other related DTCs - Harness failure - Starter motor circuit and relays circuit short circuit to ground, short circuit to power, open circuit, high resistance - Vehicle battery failure - Starter motor jammed - Starter motor mechanical failure	and check the starter motor circuit relays circuit for short circuit to ground, short circuit to power, open circuit, resistance - Refer to the workshop manual and battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests - Check starter motor for mechanical integrity - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P0A0F-71	Engine Failed to Start - Actuator stuck	NOTE: To indicate that the starter motor is too hot and is overloaded - The engine control module has not detected any motion, in response to energizing a motor, solenoid or relay - Engine control module has logged an extended crank with no engine start - Other related DTCs - Harness failure - Starter motor circuit and relays circuit	- Check engine control module for related DTCs and refer to relevant index - Refer to the electrical circuit diagram and check the starter motor circuit relays circuit for short circuit to ground, short circuit to power, open circuit, resistance - Refer to the workshop manual and battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests - Check starter motor for mechanical integrity - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		short circuit to ground, short circuit to power, open circuit, high resistance ▪ Vehicle battery failure ▪ Starter motor jammed ▪ Starter motor mechanical failure	
P0A0F-97	Engine Failed to Start - Component or system operation obstructed or blocked	NOTE: To diagnose when starter motor activation results in no engine rotation and / or safety monitoring has terminated activation ▪ The engine control module has detected that the operation of a component is prevented by an obstruction ▪ Engine control module has logged an extended crank with no engine start ▪ Other related DTCs ▪ Harness failure - Starter motor circuit and relays circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Vehicle battery	▪ Check engine control module for related DTCs and refer to relevant index ▪ Refer to the electrical circuit diagram and check the starter motor circuit relays circuit for short circuit to ground, short circuit to power, open circuit, resistance ▪ Refer to the workshop manual and battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests ▪ Check starter motor for mechanical integrity ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest

		failure ▪ Starter motor jammed ▪ Starter motor mechanical failure	
P0A0F-98	Engine Failed to Start - Component or system over temperature	▪ The engine control module has detected that the temperature is too high for the correct operation of the component or system ▪ Engine control module has logged an extended crank with no engine start ▪ Other related DTCs ▪ Harness failure - Starter motor circuit and relays circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Vehicle battery failure ▪ Starter motor jammed ▪ Starter motor mechanical failure	▪ Check engine control module for related DTCs and refer to relevant index ▪ Refer to the electrical circuit diagram and check the starter motor circuit and relays circuit for short circuit to ground, short circuit to power, open circuit, resistance ▪ Refer to the workshop manual and battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests ▪ Check starter motor for mechanical integrity ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P0A11-00	DC/DC Converter Enable Circuit/Open - No sub type information		▪ Check engine control module for related DTCs and refer to relevant index ▪ Refer to the electrical circuit diagram and check connections are secure and wiring integrity ▪ Refer to the electrical circuit diagram and check the fuel injector circuit for short circuit to power ▪ Check the vehicle charging system performance to ensure the voltage regulation is correct

		NOTE: To monitor stopping of fuel injector supply DC/DC converter bank 1/2. DC/DC buffer capacitor can not be switched off - Other related DTCs - Harness failure - Wiring integrity - Fuel injector circuit short circuit to power - Vehicle battery over charged - Charging system fault - over charging - Connector is disconnected, connector pin is backed out, connector pin corrosion - Engine control module power and ground circuits open circuit, high resistance - Engine control module mechanical integrity	- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Check engine control module for mechanical integrity - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take and retest
P0A14-13	Engine Mount Control A Circuit/Open - Circuit open	The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an	- Refer to the electrical circuit diagram and check the engine mount - Active left/right circuit for short circuit to ground, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		output ▪ Engine mount - Active, - left/right circuit short circuit to ground, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine mount - Active, - left/right failure	▪ Check and install a new engine mount - Active, - left/right as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P0A15-11	Engine Mount "A" Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Engine mount - Active, - left/right circuit short circuit to ground, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine mount - Active, - left/right failure	▪ Refer to the electrical circuit diagram and check the engine mount - Active left/right circuit for short circuit to ground, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new engine mount - Active, - left/right as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P0A16-12	Engine Mount "A" Circuit High - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle	▪ Refer to the electrical circuit diagram and check the engine mount - Active left/right circuit for short circuit to power ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		power measurement when another value was expected - Engine mount - Active, - left/right circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - Engine mount - Active, - left/right failure	- Check and install a new engine mount - Active, - left/right as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P0A16-4B	Engine Mount "A" Circuit High - Over temperature	- The engine control module detected an internal temperature above the expected range - Engine mount - Active, - left/right mechanical integrity - Engine mount - Active, - left/right circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Engine mount - Active, - left/right failure	- Check engine mount - Active, - left/right for mechanical integrity - Refer to the electrical circuit diagram and check the engine mount - Active, - left/right circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new engine mount - Active, - left/right as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P0A7B-00	Battery Energy Control Module Requested MIL Illumination - No sub type information	- MIL request by battery energy control module - HS CAN network, short circuit to	- Check battery energy control module for DTCs and refer to relevant DTC index. Rectify these first - If this DTC is logged with other HS CAN related DTCs refer to the elec

		power, short circuit to ground, open circuit Connector is disconnected, connector pin is backed out, connector pin corrosion	circuit diagrams and check HS CAN network for short circuit to power, short circuit to ground, open circuit - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P0A94-53	DC/DC Converter Performance - Deactivated	- The engine control module has indicated that some portion of the software program has been disabled - This DTC has set because the hybrid DC to DC converter has been disabled by the service routine	- Using the Jaguar Land Rover approved diagnostic equipment carry out 'Hybrid DC to DC converter mode' routine (0x4089) using the 'Service Function' tab - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis menu' tab and retest
P0AB9-68	Hybrid System Performance - Event information	- The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module - The DTC is set when the vehicle supervisory controller sends the engine control module a flag to indicate that an inertia start has been attempted. An inertia start is where vehicle supervisory controller analyses vehicle speed and	- Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first - Refer to the battery care requirements section 414-00, inspect the vehicle battery and ensure it is fully charged and serviceable - Check the vehicle charging system performance to ensure the voltage regulation is correct - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis menu' tab and retest

		gear position when driving in EV mode and decides if it is possible to bump start the engine, by closing the K0 clutch and connecting the engine to the spinning drivetrain. The only time inertia start is used is when the electric power inverter converter reports that it is no longer capable of supporting 12 volt loads. Therefore to use the starter motor to initiate a restart from EV mode in these conditions could risk the 12 volt supply dropping too low to keep all modules alive. Therefore if vehicle is moving fast enough an inertia start is performed and engine control module is told to inhibit the starter motor. If vehicle is not moving fast enough or the inertia start fails then the starter motor will be used when the vehicle comes to rest	
P115B-00	Low Fuel Level - Forced Engine Shutdown - No subtype information	▪ Low level fuel condition enabling run dry strategy ▪ Other fuel related DTCs ▪ Fuel leakage ▪ Fuel level sensor circuit short circuit to ground, short circuit to power, open circuit, high	▪ Check the fuel level, add fuel if required ▪ Check engine control module for fuel related DTCs and refer to relevant index ▪ Check fuel system for mechanical integrity ▪ Refer to the electrical circuit diagram and check the fuel level sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water

		resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel level sensor failure	ingress, and pins for damage and/corrosion - Check and install a new fuel level sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P115D-00	Mass Air Flow Circuit Offset - No sub type information	- Air flow disruption at sensing element of mass air flow sensor - Contamination at sensing element of mass air flow sensor - Mass air flow sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Mass air flow sensor failure	- Check the condition of the air filter examine the induction pipes for debris which could disrupt air flow at the sensing element - Check for contamination at sensing element of mass air flow sensor - Refer to the electrical circuit diagram and check the mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/corrosion - Check and install a new mass air flow sensor as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Engine control module mass air flow adaption clear' routine (0x0406_0E) - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Diesel fuelling adaptation re-initialization' routine (0x4020) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P115D-86	Mass Air Flow Circuit Offset - Signal invalid	The engine control module has determined failures where some circuit quantity, reported via serial data, is not plausible given the	- Check the condition of the air filter examine the induction pipes for debris which could disrupt air flow at the sensing element - Check for contamination at sensing element of mass air flow sensor

		operating conditions ▪ Air flow disruption at sensing element of mass air flow sensor ▪ Contamination at sensing element of mass air flow sensor ▪ Mass air flow sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Mass air flow sensor failure	▪ Refer to the electrical circuit diagram and check the mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new mass air flow sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out 'Engine control module mass air flow adaptation clear' routine (0x0406_0E) ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out 'Diesel fuelling adaptation re-initialization' routine (0x402C) ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P1400-11	Differential Pressure Feedback EGR Circuit Low - Circuit short to ground	NOTE: Low pressure EGR differential pressure sensor voltage signal is below a low threshold (0.2V) ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Low pressure EGR differential pressure sensor signal circuit short circuit to	▪ Diagnosis of this DTC may require using the Jaguar Land Rover approved diagnostic equipment check data log signals ▪ 0x05C7 Low Pressure EGR Pressure Sensor Voltage - Volts ▪ Refer to the electrical circuit diagram and check the low pressure EGR differential pressure sensor signal circuit for short circuit to ground ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		ground	
P1400-16	Differential Pressure Feedback EGR Circuit Low - Circuit Voltage Below Threshold	NOTE: Low pressure EGR differential pressure sensor voltage signal is below a low threshold (0.2V) - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Low pressure EGR differential pressure sensor signal circuit short circuit to ground	- Diagnosis of this DTC may require the Jaguar Land Rover approved diagnostic equipment check data logs for signals 0x05C7 Low Pressure EGR Pressure Sensor Voltage - Volts - Refer to the electrical circuit diagram and check the low pressure EGR differential pressure sensor signal circuit for short circuit to ground - Using the Jaguar Land Rover approved diagnostic equipment, perform routine diagnostic tests - Inline diagnostic unit 2 diagnostic test - Exhaust gas recirculation valve. Clear the DTCs and retest - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P1400-23	Differential Pressure Feedback EGR Circuit Low - Signal stuck low	NOTE: Low pressure EGR differential pressure sensor rationality check at initialisation, pressure below low threshold (-20hPa) - The engine control module measures a signal that remains low when transitions are expected - Low pressure EGR differential pressure sensor signal circuit short circuit to	- Diagnosis of this DTC may require the Jaguar Land Rover approved diagnostic equipment check data logs for signals 0x05C8 Low Pressure EGR Pressure 0x05C7 Low Pressure EGR Pressure Sensor Voltage - Volts - Check low pressure EGR differential pressure sensor, low pressure side hose line for - Blockage - Restricted - Refer to the electrical circuit diagram and check the low pressure EGR differential pressure sensor circuit for short circuit to ground - Using the Jaguar Land Rover approved diagnostic equipment, perform routine diagnostic tests - Inline diagnostic unit 2 diagnostic test - Exhaust gas recirculation valve. Clear the DTCs and retest

		ground - Connector is disconnected, connector pin is backed out, connector pin corrosion - Low pressure EGR differential pressure sensor failure	- Check and install a new low pressure EGR differential pressure sensor as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Air path set-up routine's (0x4030/0x4051_01/0x4050_01/0xF405) - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Engine control module low pressure EGR adaption clear routine (0x0406_0C) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P1400-84	Differential Pressure Feedback EGR Circuit Low - Signal below allowable range	NOTE: Low pressure EGR differential pressure sensor rationality check, pressure below low threshold (-20hPa) - The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range - Low pressure EGR differential pressure sensor signal circuit short circuit to ground - Connector is disconnected, connector pin is backed out, connector pin	- Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data log signals 0x05C8 Low Pressure EGR Pressure 0x05C7 Low Pressure EGR Pressure Sensor Voltage - Volts - Check low pressure EGR differential pressure sensor, low pressure side hose/line for - Blockage - Restricted - Refer to the electrical circuit diagram and check the low pressure EGR differential pressure sensor circuit for short circuit to ground - Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Inline diagnostic unit 2 diagnostic - Exhaust gas recirculation valve. Clear the DTCs and retest - Check and install a new low pressure EGR differential pressure sensor as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Air path set-up routine's (0x4030/0x4051_01/0x4050_01/0xF405)

		corrosion Low pressure EGR differential pressure sensor failure	0xF405) - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Engine control module low pressure EGR adaption clear routine (0x0406_0C) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P1401-15	Differential Pressure Feedback EGR Circuit High - Circuit short to battery or open	NOTE: Low pressure EGR differential pressure sensor voltage signal is exceeding a high threshold (4.95V) - The engine control module has detected a power measurement or open circuit for a period longer than expected or has detected a power measurement or open circuit when another value was expected - Low pressure EGR differential pressure sensor signal circuit short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Low pressure EGR differential pressure	- Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data logger signals 0x05C7 Low Pressure EGR Pressure Sensor Voltage - Volts - Refer to the electrical circuit diagram and check the low pressure EGR differential pressure sensor 5 volt supply, ground, signal circuit for short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagram and check the turbine intake shut-off valve position sensor 5 volt supply, ground circuit for open circuit, high resistance - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		sensor failure ▪ Turbine intake shut-off valve position sensor 5 volt supply, ground, circuit open circuit, high resistance	
P1401-17	Differential Pressure Feedback EGR Circuit High - Circuit voltage above threshold	NOTE: Low pressure EGR differential pressure sensor voltage signal is exceeding a high threshold (4.95V) ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Low pressure EGR differential pressure sensor signal circuit short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Low pressure EGR differential pressure sensor failure ▪ Turbine intake shut-off valve position sensor 5 volt supply, ground, circuit open circuit, high resistance	▪ Diagnosis of this DTC may require the Jaguar Land Rover approved diagnostic equipment check data signals ▪ 0x05C7 Low Pressure EGR Pressure Sensor Voltage - Volts ▪ Refer to the electrical circuit diagram and check the low pressure EGR differential pressure sensor 5 volt supply, ground, signal circuit for short circuit to power, open circuit, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform routine maintenance - Inline diagnostic unit 2 diagnostic - Exhaust gas recirculation valve. Clear the DTCs and retest ▪ Refer to the electrical circuit diagram and check the turbine intake shut-off valve position sensor 5 volt supply, ground circuit for open circuit, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest

P1401-24	Differential Pressure Feedback EGR Circuit High - Signal stuck high	NOTE: Low pressure EGR differential pressure sensor rationality check at initialisation, pressure above high threshold (20hPa) - The engine control module measures a signal that remains high when transitions are expected - Low pressure EGR differential pressure sensor signal circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - Low pressure EGR differential pressure sensor failure	- Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data for the following signals 0x05C8 Low Pressure EGR Pressure Sensor Voltage - Volts 0x05C7 Low Pressure EGR Pressure Sensor Voltage - Volts - Check low pressure EGR differential pressure sensor, high pressure side hoses for - Blockage - Restricted - Refer to the electrical circuit diagram and check the low pressure EGR differential pressure sensor circuit for short circuit to power - Using the Jaguar Land Rover approved diagnostic equipment, perform routine diagnostic tests - Inline diagnostic unit 2 diagnostic test - Exhaust gas recirculation valve. Clear the DTCs and retest - Check and install a new low pressure EGR differential pressure sensor as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Air path set-up routine' (0x4030/0x4051_01/0x4050_01/0x4050_02/0x4050_03/0x4050_04/0x4050_05/0x4050_06/0x4050_07/0x4050_08/0x4050_09/0x4050_0A/0x4050_0B/0x4050_0C/0x4050_0D/0x4050_0E/0x4050_0F/0x4050_10/0x4050_11/0x4050_12/0x4050_13/0x4050_14/0x4050_15/0x4050_16/0x4050_17/0x4050_18/0x4050_19/0x4050_1A/0x4050_1B/0x4050_1C/0x4050_1D/0x4050_1E/0x4050_1F/0x4050_20/0x4050_21/0x4050_22/0x4050_23/0x4050_24/0x4050_25/0x4050_26/0x4050_27/0x4050_28/0x4050_29/0x4050_2A/0x4050_2B/0x4050_2C/0x4050_2D/0x4050_2E/0x4050_2F/0x4050_30/0x4050_31/0x4050_32/0x4050_33/0x4050_34/0x4050_35/0x4050_36/0x4050_37/0x4050_38/0x4050_39/0x4050_3A/0x4050_3B/0x4050_3C/0x4050_3D/0x4050_3E/0x4050_3F/0x4050_40/0x4050_41/0x4050_42/0x4050_43/0x4050_44/0x4050_45/0x4050_46/0x4050_47/0x4050_48/0x4050_49/0x4050_4A/0x4050_4B/0x4050_4C/0x4050_4D/0x4050_4E/0x4050_4F/0x4050_50/0x4050_51/0x4050_52/0x4050_53/0x4050_54/0x4050_55/0x4050_56/0x4050_57/0x4050_58/0x4050_59/0x4050_5A/0x4050_5B/0x4050_5C/0x4050_5D/0x4050_5E/0x4050_5F/0x4050_60/0x4050_61/0x4050_62/0x4050_63/0x4050_64/0x4050_65/0x4050_66/0x4050_67/0x4050_68/0x4050_69/0x4050_6A/0x4050_6B/0x4050_6C/0x4050_6D/0x4050_6E/0x4050_6F/0x4050_70/0x4050_71/0x4050_72/0x4050_73/0x4050_74/0x4050_75/0x4050_76/0x4050_77/0x4050_78/0x4050_79/0x4050_7A/0x4050_7B/0x4050_7C/0x4050_7D/0x4050_7E/0x4050_7F/0x4050_80/0x4050_81/0x4050_82/0x4050_83/0x4050_84/0x4050_85/0x4050_86/0x4050_87/0x4050_88/0x4050_89/0x4050_8A/0x4050_8B/0x4050_8C/0x4050_8D/0x4050_8E/0x4050_8F/0x4050_90/0x4050_91/0x4050_92/0x4050_93/0x4050_94/0x4050_95/0x4050_96/0x4050_97/0x4050_98/0x4050_99/0x4050_9A/0x4050_9B/0x4050_9C/0x4050_9D/0x4050_9E/0x4050_9F/0x4050_A0/0x4050_A1/0x4050_A2/0x4050_A3/0x4050_A4/0x4050_A5/0x4050_A6/0x4050_A7/0x4050_A8/0x4050_A9/0x4050_AA/0x4050_AB/0x4050_AC/0x4050_AD/0x4050_AE/0x4050_AF/0x4050_B0/0x4050_B1/0x4050_B2/0x4050_B3/0x4050_B4/0x4050_B5/0x4050_B6/0x4050_B7/0x4050_B8/0x4050_B9/0x4050_BA/0x4050_BB/0x4050_BC/0x4050_BD/0x4050_BE/0x4050_BF/0x4050_C0/0x4050_C1/0x4050_C2/0x4050_C3/0x4050_C4/0x4050_C5/0x4050_C6/0x4050_C7/0x4050_C8/0x4050_C9/0x4050_CA/0x4050_CB/0x4050_CC/0x4050_CD/0x4050_CE/0x4050_CF/0x4050_D0/0x4050_D1/0x4050_D2/0x4050_D3/0x4050_D4/0x4050_D5/0x4050_D6/0x4050_D7/0x4050_D8/0x4050_D9/0x4050_DA/0x4050_DB/0x4050_DC/0x4050_DD/0x4050_DE/0x4050_DF/0x4050_E0/0x4050_E1/0x4050_E2/0x4050_E3/0x4050_E4/0x4050_E5/0x4050_E6/0x4050_E7/0x4050_E8/0x4050_E9/0x4050_EA/0x4050_EB/0x4050_EC/0x4050_ED/0x4050_EE/0x4050_EF/0x4050_F0/0x4050_F1/0x4050_F2/0x4050_F3/0x4050_F4/0x4050_F5/0x4050_F6/0x4050_F7/0x4050_F8/0x4050_F9/0x4050_FA/0x4050_FB/0x4050_FC/0x4050_FD/0x4050_FE/0x4050_FF) - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Engine control module adaptation' (0x4046_0C) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P1401-85	Differential Pressure Feedback EGR Circuit High - Signal Above Allowable Range		- Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data for the following signals 0x05C8 Low Pressure EGR Pressure Sensor Voltage - Volts 0x05C7 Low Pressure EGR Pressure Sensor Voltage - Volts

		NOTE: Low pressure EGR loop - Low pressure EGR differential pressure sensor, pressure rationality check; pressure above high threshold (375hPa) ▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range ▪ Low pressure EGR differential pressure sensor, high pressure side hoseline ▪ Blocked ▪ Restricted ▪ Low pressure EGR differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Check low pressure EGR differential pressure sensor, high pressure side hoseline for ▪ Blockage ▪ Restricted ▪ Refer to the electrical circuit diagram and check the low pressure EGR differential pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform routine diagnostic tests - Inline diagnostic unit 2 diagnostic tests - Exhaust gas recirculation valve. Clear the DTCs and retest ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P1405-00	Differential Pressure Feedback Sensor Upstream Hose Off Or Plugged - No sub type information		▪ Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data logger signals ▪ 0x05C8 Low Pressure EGR Pressure

		NOTE: To monitor low pressure EGR differential pressure sensor, high pressure side hoseline is installed - Low pressure EGR differential pressure sensor, high pressure side hoseline - Not installed - Disconnected - Damaged (leakage) - Exhaust gas recirculation - Low pressure butterfly valve stuck closed	- Check low pressure EGR differential pressure sensor, high pressure side hoseline for - Correct installation - Damaged (leakage) - Check exhaust gas recirculation - Low pressure butterfly valve for mechanical integrity - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P1551-32	Cylinder 1 Injector Circuit Range/Performance - Signal low time < minimum	- The engine control module detected the low pulse is too narrow with respect to time - Other fuel injector related DTCs - Fuel injector is mechanically degraded - Fuel injector is carbonized	- Check engine control module for fuel injection related DTCs and refer to relevant DTC index - Check fuel injector for mechanical integrity - Check and install a new fuel injector if required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Injector replacement' routine - Using the Jaguar Land Rover approved diagnostic equipment reprogram the injector codes - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P1551-35	Cylinder 1 Injector Circuit Range/Performance -	This DTC is set when the engine control	Check engine control module for fuel injection related DTCs and refer to

	Signal high time > maximum	module detects that the signal high time is greater than the maximum value - Other fuel injector related DTCs - Fuel injector is mechanically degraded - Fuel injector is carbonized	relevant DTC index - Check fuel injector for mechanical integrity - Check and install a new fuel injector required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Injector replacement' routine - Using the Jaguar Land Rover approved diagnostic equipment reprogram the injector codes - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P1552-32	Cylinder 2 Injector Circuit Range/Performance - Signal low time < minimum	- The engine control module detected the low pulse is too narrow with respect to time - Other fuel injector related DTCs - Fuel injector is mechanically degraded - Fuel injector is carbonized	- Check engine control module for fuel injection related DTCs and refer to relevant DTC index - Check fuel injector for mechanical integrity - Check and install a new fuel injector required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Injector replacement' routine - Using the Jaguar Land Rover approved diagnostic equipment reprogram the injector codes - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P1552-35	Cylinder 2 Injector Circuit Range/Performance - Signal high time > maximum	- This DTC is set when the engine control module detects that the signal high time is greater than the maximum value - Other fuel injector related DTCs - Fuel injector is mechanically	- Check engine control module for fuel injection related DTCs and refer to relevant DTC index - Check fuel injector for mechanical integrity - Check and install a new fuel injector required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Injector replacement'

		degraded Fuel injector is carbonized	routine - Using the Jaguar Land Rover approved diagnostic equipment reprogram the injector codes - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P1553-32	Cylinder 3 Injector Circuit Range/Performance - Signal low time < minimum	- The engine control module detected the low pulse is too narrow with respect to time - Other fuel injector related DTCs - Fuel injector is mechanically degraded - Fuel injector is carbonized	- Check engine control module for fuel injection related DTCs and refer to relevant DTC index - Check fuel injector for mechanical integrity - Check and install a new fuel injector required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Injector replacement' routine - Using the Jaguar Land Rover approved diagnostic equipment reprogram the injector codes - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P1553-35	Cylinder 3 Injector Circuit Range/Performance - Signal high time > maximum	- This DTC is set when the engine control module detects that the signal high time is greater than the maximum value - Other fuel injector related DTCs - Fuel injector is mechanically degraded - Fuel injector is carbonized	- Check engine control module for fuel injection related DTCs and refer to relevant DTC index - Check fuel injector for mechanical integrity - Check and install a new fuel injector required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Injector replacement' routine - Using the Jaguar Land Rover approved diagnostic equipment reprogram the injector codes - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

P1554-32	Cylinder 4 Injector Circuit Range/Performance - Signal low time < minimum	▪ The engine control module detected the low pulse is too narrow with respect to time ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injector required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out 'Injector replacement' routine ▪ Using the Jaguar Land Rover approved diagnostic equipment reprogram the injector codes ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take and retest
P1554-35	Cylinder 4 Injector Circuit Range/Performance - Signal high time > maximum	▪ This DTC is set when the engine control module detects that the signal high time is greater than the maximum value ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injector required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out 'Injector replacement' routine ▪ Using the Jaguar Land Rover approved diagnostic equipment reprogram the injector codes ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take and retest
P1555-32	Cylinder 5 Injector Circuit Range/Performance - Signal low time < minimum	▪ The engine control module detected the low pulse is too narrow with respect to time ▪ Other fuel injector related DTCs	▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injector required

		▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Using the Jaguar Land Rover approved diagnostic equipment carry out 'Injector replacement' routine ▪ Using the Jaguar Land Rover approved diagnostic equipment reprogram the injector codes ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take and retest
P1555-35	Cylinder 5 Injector Circuit Range/Performance - Signal high time > maximum	▪ This DTC is set when the engine control module detects that the signal high time is greater than the maximum value ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injector required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out 'Injector replacement' routine ▪ Using the Jaguar Land Rover approved diagnostic equipment reprogram the injector codes ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take and retest
P1556-32	Cylinder 6 Injector Circuit Range/Performance - Signal low time < minimum	▪ The engine control module detected the low pulse is too narrow with respect to time ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injector required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out 'Injector replacement' routine ▪ Using the Jaguar Land Rover approved diagnostic equipment reprogram the injector codes ▪ Using the Jaguar Land Rover approved

			diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P1556-35	Cylinder 6 Injector Circuit Range/Performance - Signal high time > maximum	▪ This DTC is set when the engine control module detects that the signal high time is greater than the maximum value ▪ Other fuel injector related DTCs ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Check and install a new fuel injector if required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out 'Injector replacement' routine ▪ Using the Jaguar Land Rover approved diagnostic equipment reprogram the injector codes ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P1592-44	Vehicle Data Recorder Data Available - Data memory failure	NOTE: Calibration error, data corruption during write to non volatile memory (battery disconnect during engine control module write sequence), internal engine control module error ▪ Incorrect engine control module calibration for vehicle specification ▪ Engine control	▪ Using the Jaguar Land Rover approved diagnostic equipment check and install the latest relevant level of software to the engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance ▪ Check engine control module for mechanical integrity ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest

		module software calibration error ▪ Engine control module power and ground circuits open circuit, high resistance ▪ Engine control module mechanical integrity ▪ Corrupt engine control module software download	
P1592-46	Vehicle Data Recorder Data Available - Calibration / parameter memory failure	NOTE: Calibration error, data corruption during write to non volatile memory (battery disconnect during engine control module write sequence), internal engine control module error ▪ Incorrect engine control module calibration for vehicle specification ▪ Engine control module software calibration error ▪ Engine control module power and ground circuits open circuit, high resistance ▪ Engine control module mechanical integrity	▪ Using the Jaguar Land Rover approved diagnostic equipment check and install the latest relevant level of software to the engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance ▪ Check engine control module for mechanical integrity ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest

		Corrupt engine control module software download	
P162F-00	Starter Motor Disabled - Engine Crank Time Too Long - No sub type information	NOTE: To inform that auto stop/start system was deactivated as the engine failed to restart in the required time - Other starting related DTCs - Other fuelling related DTCs - Starter motor mechanical integrity - Vehicle battery failure	- Check engine control module for starting related DTCs and refer to relevant DTC index - Check engine control module for fuelling related DTCs and refer to relevant DTC index - Check starter motor for mechanical integrity - Refer to the workshop manual and battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P16A3-00	SCR System - EOL - Failure During Emptying Test - No sub type information	- Purpose of the DTC - Selective catalytic reduction dosing system prime routine failed during the empty phase - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Not applicable - Monitor Description - Selective catalytic reduction dosing system prime routine failed	- Vehicle Conditions to enable DTC Logging strategy - Ignition on - Prioritised Checks to Perform - Diagnosis of this DTC may require using the Jaguar Land Rover approved diagnostic equipment check datalogger signals 0x05C1 Reductant Tank Fluid Level 0x043C Reductant Tank Temperature - Raw - Check that the DEF tank has at least 25mm of fluid (0x05C1) - Check the DEF tank temperature within -5 to + 70 °C (0x043C) - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high

		during the empty phase ▪ Prioritised List of Possible Causes ▪ Selective catalytic reduction priming routine did not complete successfully ▪ DEF back flow pump did not empty the pressure line correctly ▪ DEF pressure line leakage ▪ Routine aborted at ignition off	resistance ▪ Check the DEF pressure line for leakage ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine 0x409A - Diesel Exhaust Fluid Prime and Pressure Test. Check that the ignition stays on throughout the whole duration of the routine ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P16A4-00	SCR System - EOL - Failure During First Fill Test - No sub type information	▪ Purpose of the DTC ▪ Selective catalytic reduction dosing system prime routine failed during the first fill phase ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Selective catalytic reduction dosing system prime routine failed during the first fill phase ▪ Prioritised List of Possible Causes ▪ Selective catalytic	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Ignition on ▪ Prioritised Checks to Perform ▪ Diagnosis of this DTC may require using the Jaguar Land Rover approved diagnostic equipment check datalogger signals ▪ 0x05C1 Reductant Tank Fluid Level ▪ 0x043C Reductant Tank Temperature - Raw ▪ Check that the DEF tank has at least 25mm of fluid (0x05C1) ▪ Check the DEF tank temperature within -5 to + 70 °C (0x043C) ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check the DEF pressure line for leakage ▪ Using the Jaguar Land Rover approved diagnostic equipment

		reduction priming routine did not complete successfully - Pressure and DEF back flow pump not activated correctly - Routine aborted at ignition off	carry out routine 0x409A - Dies Exhaust Fluid Prime and Pressu Test. Check that the ignition sta on throughout the whole durati of the routine Using the Jaguar Land Rover approved diagnostic equipmen clear all stored DTCs using the Diagnosis Menu tab and retest
P16A5-00	SCR System - EOL - Failure During Pressure Test - No sub type information	- Purpose of the DTC - Selective catalytic reduction dosing system prime routine failed during the pressure preparation phase - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Not applicable - Monitor Description - Selective catalytic reduction dosing system prime routine failed during the pressure preparation phase - Prioritised List of Possible Causes - Selective catalytic reduction priming routine did not complete successfully	- Vehicle Conditions to enable DTC Logging strategy - Ignition on - Prioritised Checks to Perform - Diagnosis of this DTC may requ using the Jaguar Land Rover approved diagnostic equipmen check datalogger signals 0x05C1 Reductant Tank Fluid L 0x043C Reductant Tank Temperature - Raw - Check that the DEF tank has at least 25mm of fluid (0x05C1) - Check the DEF tank temperatu within -5 to + 70 °C (0x043C) - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circu short circuit to ground, short ci to power, open circuit, high resistance - Check the DEF pressure line fo leakage - Using the Jaguar Land Rover approved diagnostic equipmen carry out routine 0x409A - Dies Exhaust Fluid Prime and Pressu Test. Check that the ignition sta on throughout the whole durati of the routine - Using the Jaguar Land Rover approved diagnostic equipmen clear all stored DTCs using the Diagnosis Menu tab and retest

		- Pressure pump is not pressurizing the line correctly - DEF pressure line leakage - Routine aborted at ignition off	
P16A6-00	SCR System - EOL - Unfinished Tests - No sub type information	- Purpose of the DTC - Selective catalytic reduction dosing system prime routine aborted during cycle - Electrical Cause - No - Mechanical Cause - No - Control Module Cavity - Not applicable - Monitor Description - Selective catalytic reduction dosing system prime routine aborted during cycle - Prioritised List of Possible Causes - Selective catalytic reduction priming routine did not complete successfully - The routine has been aborted by the user - Routine aborted at ignition off	- Vehicle Conditions to enable DTC Logging strategy - Ignition on - Prioritised Checks to Perform - Diagnosis of this DTC may require using the Jaguar Land Rover approved diagnostic equipment check datalogger signals 0x05C1 Reductant Tank Fluid Level 0x043C Reductant Tank Temperature - Raw - Check that the DEF tank has at least 25mm of fluid (0x05C1) - Check the DEF tank temperature within -5 to + 70 °C (0x043C) - Using the Jaguar Land Rover approved diagnostic equipment carry out routine 0x409A - Diesel Exhaust Fluid Prime and Pressure Test. Check that the ignition stays on throughout the whole duration of the routine - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P16A7-00	SCR System - Tampering At Connector 1 - No sub type information	- Purpose of the DTC - Diesel exhaust fluid tank	

				module connector monitoring (pump) - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Not applicable - Monitor Description - Diesel exhaust fluid tank module (pump) connector is removed - Prioritised List of Possible Causes - Diesel exhaust fluid tank module connector is disconnected, connector pin is backed out, connector pin corrosion	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The electrical checks normally occur once ignition is on or after engine start - Prioritised Checks to Perform - Inspect connectors for signs of water ingress, and pins for dam and/or corrosion - Vehicle Conditions to enable DTC Logging strategy - Engine running - On completion of the previous repair actions the following MU be carried out - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest - NOTE: The engine MIL is OFF (when engine ON) - Start engine - The 'inducement' warning clear automatically once the monitor has executed and passed with no malfunction. This terminates the drive cycle - NOTE: If the engine MIL shows during the driving cycle: the malfunction has re-occurred. Further repair is required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab - NOTE: This cleans up the fault storage from P2BAE/P2BAF
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P16A8-00	SCR System - Tampering At Connector 2 - No sub type information	▪ Purpose of the DTC ▪ Diesel exhaust fluid tank module connector monitoring (heater/level) ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid tank module connector is removed ▪ Prioritised List of Possible Causes ▪ Diesel exhaust fluid tank module connector is disconnected, connector pin is backed out, connector pin corrosion	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The electrical checks normally occur once ignition is on or after engine start ▪ Prioritised Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Vehicle Conditions to enable DTC Logging strategy ▪ Engine running ▪ On completion of the previous repair actions the following MU be carried out ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest ▪ NOTE: The engine MIL is OFF (when engine ON) ▪ Start engine ▪ The 'inducement' warning clear automatically once the monitor has executed and passed with no malfunction. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle: the malfunction has re-occurred. Further repair is required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This cleans up the fault
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			storage from P2BAE/P2BAF
P2002-00	Diesel Particulate Filter Efficiency Below Threshold (Bank 1) - No sub type information	- Exhaust system leakage - Diesel particulate filter degraded - In cold climates differential pressure sensor hose lines or metal pipes may be frozen - Differential pressure sensor hoses connected incorrectly - Differential pressure sensor hoses crushed, blocked, split - Differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Differential pressure sensor failure	- Diagnosis of this DTC may require the Jaguar Land Rover approved diagnostic equipment check data logs signals 0xD922 Particulate Filter Differential Pressure - Measured - Measure with engine off - Check differential pressure sensor hoses lines are not frozen - Check differential pressure sensor hoses lines are installed correctly - Check differential pressure sensor hoses lines for mechanical integrity - Refer to the electrical circuit diagram and check the differential pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new differential pressure sensor as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Diesel particulate filter differential pressure sensor replacement' routine (0x405B_) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P202A-13	Reductant Tank Heater Control Circuit / Open - Circuit Open	- Purpose of the DTC - To monitor diesel exhaust fluid heater for open circuit, high resistance - Electrical Cause - Yes - Mechanical Cause - No - Control Module	- Vehicle Conditions to enable DTC Logging strategy - Engine running - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage

		- Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid heater open circuit, high resistance ▪ Prioritised List of Possible Causes ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Diesel exhaust fluid heater control unit circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P202B-00	Reductant Tank Heater Control Circuit Low - No sub type information	▪ Diesel exhaust fluid heater control unit circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit	▪ Refer to the electrical circuit diagram and check the diesel exhaust fluid heater control unit circuit short circuit to ground ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored

		failure	DTCs using the Diagnosis Menu tab and retest
P202B-11	Reductant Tank Heater Control Circuit Low - Circuit short to ground	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid heater for short circuit to ground ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid heater short circuit to ground ▪ Prioritised List of Possible Causes ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Diesel exhaust fluid heater control unit circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit short circuit to ground ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		▪ Diesel exhaust fluid heater control unit failure	
P202B-13	Reductant Tank Heater Control Circuit Low - Circuit open	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid heater for open circuit, high resistance ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid heater open circuit, high resistance ▪ Prioritised List of Possible Causes ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Diesel exhaust fluid heater control unit circuit open circuit, high resistance ▪ CAN communication diesel exhaust fluid tank module - DEF heater controller circuit open	▪ Vehicle Conditions to enable DTC - Logging strategy ▪ DEF heater on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the CAN communication diesel exhaust fluid tank module - DEF heater control circuit for open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit if required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Diesel exhaust fluid heater control unit failure	
P202C-00	Reductant Tank Heater Control Circuit High - No sub type information	- Diesel exhaust fluid heater control unit circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - Diesel exhaust fluid heater control unit failure	- Refer to the electrical circuit diagram and check the diesel exhaust fluid heater control unit circuit short circuit to power - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new diesel exhaust fluid heater control unit as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P202C-12	Reductant Tank Heater Control Circuit High - Circuit short to battery	- Purpose of the DTC - To monitor diesel exhaust fluid heater for short circuit to power - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Not applicable - Monitor Description - Diesel exhaust fluid heater short circuit to power - Prioritised List of	- Vehicle Conditions to enable DTC Logging strategy - Engine running - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit short circuit to power - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new diesel exhaust fluid heater control unit as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		Possible Causes ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Diesel exhaust fluid heater control unit circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	
P202C-15	Reductant Tank Heater Control Circuit High - Circuit short to battery or open	▪ Purpose of the DTC ▪ To monitor DEF tank heater element ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ DEF tank heater element short circuit ▪ Prioritised List of Possible Causes	▪ Vehicle Conditions to enable DTC Logging strategy ▪ DEF heater on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit short circuit to ground short circuit to power ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit if required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the

		▪ The engine control module has detected a power measurement or open circuit for a period longer than expected or has detected a power measurement or open circuit when another value was expected ▪ Diesel exhaust fluid heater control unit circuit short circuit to ground short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	Diagnosis Menu tab and retest
P202E-00	Reductant Injection Valve Circuit Range/Performance (Bank 1 Unit 1) - No sub type information	▪ Diesel exhaust fluid heater control unit circuit open circuit, high resistance ▪ CAN communication diesel exhaust fluid tank module - DEF heater controller circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Refer to the electrical circuit diagram and check the diesel exhaust fluid heater control unit circuit for open circuit, high resistance ▪ Refer to the electrical circuit diagram and check the CAN communication diesel exhaust fluid tank module - DEF heater controller circuit for open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored

		▪ Diesel exhaust fluid heater control unit failure	DTCs using the Diagnosis Menu tab and retest
P202E-4B	Reductant Injection Valve Circuit Range/Performance (Bank 1 Unit 1) - Over temperature	▪ Purpose of the DTC ▪ To monitor DEF injector for over temperature ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference RAMV_L ▪ Circuit reference RAMV_H ▪ Monitor Description ▪ DEF injector over temperature ▪ Prioritised List of Possible Causes ▪ The engine control module detected an internal temperature above the expected range ▪ Diesel exhaust fluid injector circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Diesel exhaust fluid injector overheated ▪ Diesel exhaust fluid injector failure	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check diesel exhaust fluid injector for overheating condition ▪ Check and install a new diesel exhaust fluid injector as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2031-	Exhaust Gas Temperature		▪ Diagnosis of this DTC may require

22	Sensor Circuit Bank 1 Sensor 2 - Signal amplitude > maximum	NOTE: To monitor for too high temperature at the diesel oxidation catalyst outlet location; DOC/DPF component protection. Temperature sensor temperature above threshold (874.96°C) ▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high ▪ Overfilled DPF during drop to idle with active DPF regeneration on ▪ Blocked front face of DPF ▪ Stuck closed electric throttle butterfly valve during active DPF regeneration ▪ Other related DTCs ▪ Airpath ▪ Turbocharger ▪ EGR ▪ Fuel system ▪ Exhaust gas temperature sensor - Pre catalytic converter circuit short circuit to ground, short circuit	the Jaguar Land Rover approved diagnostic equipment check data logs signals ▪ 0x03F7 Exhaust Gas Temperature Bank2 Sensor 1 ▪ 0x03E7 Exhaust Gas Temperature Bank2 Sensor 1 Voltage ▪ 0x042C Diesel Particulate Filter Soot Concentration ▪ Check DPF for mechanical integrity ▪ Check electric throttle butterfly valve for mechanical integrity ▪ Check engine control module for related DTCs and refer to relevant index ▪ Refer to the electrical circuit diagram and check the exhaust gas temperature sensor - Pre catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check and install a new exhaust gas temperature sensor - Pre catalytic converter as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
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		to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas temperature sensor - Pre catalytic converter failure	
P2032-16	Exhaust Gas Temperature Sensor Circuit Low Bank 1 Sensor 2 - Circuit voltage below threshold	NOTE: Exhaust gas temperature sensor voltage below threshold (80mV) - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Exhaust gas temperature sensor - Pre catalytic converter circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas temperature sensor -	- Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data link signals 0x03E7 Exhaust Gas Temperature Bank2 Sensor 1 Voltage - Refer to the electrical circuit diagram and check the exhaust gas temperature sensor - Pre catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check and install a new exhaust gas temperature sensor - Pre catalytic converter as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest

		Pre catalytic converter failure	
P2033-17	Exhaust Gas Temperature Sensor Circuit High Bank 1 Sensor 2 - Circuit voltage above threshold	NOTE: Exhaust gas temperature sensor voltage above threshold (4990mV) - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Exhaust gas temperature sensor - Pre catalytic converter circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas temperature sensor - Pre catalytic converter failure	- Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data for sensor signals 0x03E7 Exhaust Gas Temperature Bank2 Sensor 1 Voltage - Refer to the electrical circuit diagram and check the exhaust gas temperature sensor - Pre catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check and install a new exhaust gas temperature sensor - Pre catalytic converter as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P203B-02	Reductant Level Sensor "A" Circuit Range/Performance - General signal failure	- Purpose of the DTC - Monitoring of the diesel exhaust fluid level sensor (level) - Electrical Cause - Yes	- Vehicle Conditions to enable DTC Logging strategy - Engine running - Tank unfrozen, drive for 10 Km - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel

		▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference I_T_RAULS Signal ▪ Monitor Description ▪ Diesel exhaust fluid tank module level reading not possible ▪ Prioritised List of Possible Causes ▪ Diesel exhaust fluid tank module communication error ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P203B-16	Reductant Level Sensor "A" Circuit Range/Performance - Circuit voltage below threshold	▪ Purpose of the DTC ▪ Monitoring of the diesel exhaust fluid level sensor (level) ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		I_T_RAULS Signal - Monitor Description - Diesel exhaust fluid tank module level signal range check too low - Prioritised List of Possible Causes - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Contact dealer technical support - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P203B-17	Reductant Level Sensor "A" Circuit Range/Performance - Circuit voltage above threshold	- Purpose of the DTC - Monitoring of the diesel exhaust fluid level sensor (level) - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference I_T_RAULS Signal	- Vehicle Conditions to enable DTC Logging strategy - Engine running - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Contact dealer technical support

		- Monitor Description - Diesel exhaust fluid tank module level signal range check too high - Prioritised List of Possible Causes - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P203B-31	Reductant Level Sensor "A" Circuit Range/Performance - No signal	- Purpose of the DTC - Monitoring of the diesel exhaust fluid level sensor (level) - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference I_T_RAULS Signal - Monitor Description - DEF level value	- Vehicle Conditions to enable DTC Logging strategy - DEF tank unfrozen - Drive 10 Km - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Contact dealer technical support - Using the Jaguar Land Rover

		not available ▪ Prioritised List of Possible Causes ▪ The engine control module does not detect a signal which ought to be present ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P203C-00	Reductant Level Sensor "A" Circuit Low - No sub type information	▪ Purpose of the DTC ▪ Monitoring of the diesel exhaust fluid level sensor (level) ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference I_T_RAULS Signal ▪ Monitor Description ▪ Diesel exhaust fluid tank module level sensor low level line error ▪ Prioritised List of Possible Causes ▪ Diesel exhaust	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The electrical checks normally occur once ignition is on or after engine start ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		- fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Contact dealer technical support - Vehicle Conditions to enable DTC Logging Strategy - Engine running - On completion of the previous repair actions the following MU be carried out - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest - NOTE: The engine MIL is OFF (when engine ON) - Start engine - The 'inducement' warning clear automatically once the monitor has executed and passed with no malfunction. This terminates the drive cycle - NOTE: If the engine MIL shows during the driving cycle, then the malfunction has re-occurred. Further repair is required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab - NOTE: This clears the fault stored from P2BAE / P2BAF
P203D-00	Reductant Level Sensor "A" Circuit High - No subtype information	- Purpose of the DTC - Monitoring of the diesel exhaust fluid level sensor (level) - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference I_T_RAULS Signal - Monitor Description	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The electrical checks normally occur once ignition is on or after engine start Prioritised Checks to Perform

		▪ Diesel exhaust fluid tank module level sensor high level line error ▪ Prioritised List of Possible Causes ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ On completion of the previous repair actions the following MU be carried out ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest ▪ NOTE: The engine MIL is OFF (when engine ON) ▪ Start engine ▪ The 'inducement' warning clear automatically once the monitor has executed and passed with no malfunction. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle, then the malfunction has re-occurred. Further repair is required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This clears the fault stored from P2BAE / P2BAF
P2043-00	Reductant Temperature Sensor Circuit Range/Performance - No sub type information	▪ The engine control module detected plausibility failures ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit	▪ Refer to the electrical circuit diagram and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or

		to power, open circuit, high resistance Connector is disconnected, connector pin is backed out, connector pin corrosion	corrosion - Contact dealer technical support - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take test and retest
P2044-00	Reductant Temperature Sensor Circuit Low - No sub type information	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Diesel exhaust fluid tank module circuit short circuit to ground, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Refer to the electrical circuit diagram and check the diesel exhaust fluid tank module circuit for short circuit to ground, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Contact dealer technical support - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take test and retest
P2045-00	Reductant Temperature Sensor Circuit High - No sub type information	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Diesel exhaust fluid tank module circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Refer to the electrical circuit diagram and check the diesel exhaust fluid tank module circuit for short circuit to power - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Contact dealer technical support - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take test and retest
P2047-	Reductant Injection Valve	Purpose of the DTC	

- To monitor DEF injector circuit for open circuit
- Electrical Cause
 - Yes
- Mechanical Cause
 - No
- Control Module Cavity
 - Circuit reference RAMV_L
 - Circuit reference RAMV_H
- Monitor Description
 - DEF injector circuit open circuit, high resistance
- Prioritised List of Possible Causes
 - The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output
 - Diesel exhaust fluid injector circuit open circuit, high resistance
 - Connector is disconnected, connector pin is backed out, connector pin corrosion
 - Diesel exhaust fluid injector failure

NOTE:

A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The electrical checks normally occur once ignition is on or after engine start

- Prioritised Checks to Perform
 - Refer to the electrical circuit diagrams and check the diesel exhaust fluid injector circuit for open circuit, high resistance
 - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
 - Check and install a new diesel exhaust fluid injector as required
- Vehicle Conditions to enable DTC Logging Strategy
 - Engine running
 - On completion of the previous repair actions the following MU be carried out
 - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
 - NOTE: The engine MIL is OFF (when engine ON)
 - Start engine
 - The 'inducement' warning clear automatically once the monitor has executed and passed with no malfunction. This terminates the drive cycle
 - NOTE: If the engine MIL shows during the driving cycle, then the malfunction has re-occurred. Further repair is required

			- Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab - NOTE: This clears the fault store from P2BAE / P2BAF
P2048-00	Reductant Injection Valve Circuit Low (Bank 1 Unit 1) - No sub type information	- Diesel exhaust fluid injector circuit short circuit to ground - Connector is disconnected, connector pin is backed out, connector pin corrosion - Diesel exhaust fluid injector failure	- Refer to the electrical circuit diagram and check the diesel exhaust fluid injector circuit for short circuit to ground - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new diesel exhaust fluid injector as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2048-11	Reductant Injection Valve Circuit Low (Bank 1 Unit 1) - Circuit short to ground	- Purpose of the DTC - To monitor DEF injector circuit for short circuit to ground - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference RAMV_L - Circuit reference RAMV_H - Monitor Description - DEF injector circuit for short circuit to ground - Prioritised List of Possible Causes - The engine control module has detected a ground	- Vehicle Conditions to enable DTC Logging strategy - Engine running - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid injector circuit for short circuit to ground - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new diesel exhaust fluid injector as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Diesel exhaust fluid injector circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid injector failure	
P2049-00	Reductant Injection Valve Circuit High (Bank 1 Unit 1) - No sub type information	▪ Diesel exhaust fluid injector circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid injector failure	▪ Refer to the electrical circuit diagram and check the diesel exhaust fluid injector circuit for short circuit to power ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid injector as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2049-12	Reductant Injection Valve Circuit High (Bank 1 Unit 1) - Circuit short to battery	▪ Purpose of the DTC ▪ To monitor DEF injector circuit for short circuit to power ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid injector circuit for short circuit to power ▪ Inspect connectors for signs of water ingress, and pins for damage

		Cavity ▪ Circuit reference RAMV_L ▪ Circuit reference RAMV_H ▪ Monitor Description ▪ DEF injector circuit for short circuit to ground ▪ Prioritised List of Possible Causes ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Diesel exhaust fluid injector circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid injector failure	and/or corrosion ▪ Check and install a new diesel exhaust fluid injector as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P205B-84	Reductant Tank Temperature Sensor Circuit Range/Performance - Signal below allowable range	▪ Purpose of the DTC ▪ Monitoring of the diesel exhaust fluid level sensor (temperature) ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Cold start detection (engine off more than 28000 seconds) ▪ Cold start detection (engine off more than 28000s, Tenv Tfuel and Tcat within 7°C) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit

		▪ Yes ▪ Control Module Cavity ▪ Circuit reference I_T_RAULS Signal ▪ Monitor Description ▪ Temperature difference below maximum limit ▪ Prioritised List of Possible Causes ▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P205B-85	Reductant Tank Temperature Sensor Circuit Range/Performance - Signal above allowable range	▪ Purpose of the DTC ▪ Monitoring of the diesel exhaust fluid level sensor (temperature) ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Cold start detection (engine off more than 28000 seconds) ▪ Cold start detection (engine off more than 28000s, Tenv Tfuel and Tcat within 7°C) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance

		- Cavity - Circuit reference I_T_RAULS Signal - Monitor Description - Temperature difference above maximum limit - Prioritised List of Possible Causes - The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range - Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	- to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Contact dealer technical support - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P205C-16	Reductant Tank Temperature Sensor Circuit Low - Circuit voltage below threshold	- Purpose of the DTC - Monitoring of the diesel exhaust fluid level sensor (temperature) - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference I_T_RAULS	- Vehicle Conditions to enable DTC Logging strategy - Engine running - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Contact dealer technical support

		Signal - Monitor Description - Diesel exhaust fluid tank module temperature signal range check too low - Prioritised List of Possible Causes - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P205D-17	Reductant Tank Temperature Sensor Circuit High - Circuit voltage above threshold	- Purpose of the DTC - Monitoring of the diesel exhaust fluid level sensor (temperature) - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference I_T_RAULS Signal	- Vehicle Conditions to enable DTC Logging strategy - Engine running - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Contact dealer technical support

		- Monitor Description - Diesel exhaust fluid tank module temperature signal range check too high - Prioritised List of Possible Causes - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P205E-64	Reductant Tank Temperature Sensor Circuit Intermittent/Erratic - Signal plausibility failure	- Purpose of the DTC - Monitoring of the diesel exhaust fluid level sensor (temperature) - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference I_T_RAULS Signal - Monitor Description	- Vehicle Conditions to enable DTC Logging strategy - Key On - DEF level greater than 15% - DEF tank unfrozen - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		▪ Diesel exhaust fluid tank module temperature plausibility, signal too low ▪ Prioritised List of Possible Causes ▪ The engine control module detected plausibility failures ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Contact dealer technical support ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2080-62	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 1 - Signal compare failure	NOTE: Circuit reference - TIT_1 ▪ The engine control module detected failure when comparing two or more input parameters for plausibility ▪ Exhaust gas temperature sensor - Turbine temperature sensor circuit short circuit to ground, short circuit to power, open circuit,	NOTE: Exhaust gas temperature sensor - Turbine temperature sensor rationality check. Temperature sensor is checked against other temperature sensors at coldstart ▪ Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data for the following signals ▪ 0x05CA Exhaust Turbine A Inlet Temperature ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new exhaust gas temperature sensor - Turbine

		high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas temperature sensor - Turbine temperature sensor failure	temperature sensor as required Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take test and retest
P2080-64	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 1 - Signal plausibility failure	NOTE: Circuit reference - TIT_1 - The engine control module detected plausibility failures - Other related DTCs - Airpath - Manifold - Fuel system - Exhaust gas temperature sensor - Turbine temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas temperature sensor - Turbine temperature sensor failure	NOTE: Exhaust gas temperature sensor - Turbine temperature sensor rationality check during a driving cycle is not plausible (compared against a temperature model). Exhaust temperature sensor temperature is checked during a vehicle driving cycle against a model temperature for rationality. Some engine speed/load conditions apply. If the difference between the sensor temperature and the model exceeds a defined threshold, a malfunction is raised - Check engine control module for related DTCs and refer to relevant index - Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data/learn signals 0x05CA Exhaust Turbine A Inlet Temperature - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new exhaust gas temperature sensor - Turbine temperature sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take test and retest

			and retest
P2084-62	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 2 - Signal compare failure	NOTE: Exhaust gas temperature sensor temperature at least 50degC above other exhaust gas temperature sensors temperature after 90 minutes engine off time ▪ The engine control module detected failure when comparing two or more input parameters for plausibility ▪ Exhaust gas temperature sensor - Pre catalytic converter circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Pre catalytic converter failure	▪ Diagnosis of this DTC may require the Jaguar Land Rover approved diagnostic equipment check data signals ▪ 0x03F7 Exhaust Gas Temperature Bank2 Sensor 1 ▪ 0x03E7 Exhaust Gas Temperature Bank2 Sensor 1 Voltage ▪ Refer to the electrical circuit diagram and check the exhaust gas temperature sensor - Pre catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check and install a new exhaust gas temperature sensor - Pre catalytic converter as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2084-64	Exhaust Gas Temperature Sensor Circuit		▪ Diagnosis of this DTC may require the Jaguar Land Rover approved

Range/Performance Bank 1
Sensor 2 - Signal
plausibility failure

NOTE:

Exhaust gas temperature sensor temperature is checked during a vehicle driving cycle against a model temperature for rationality. Some engine speed/load conditions apply. If the difference between the sensor temperature and the model exceeds a defined threshold, the DTC is set

- The engine control module detected plausibility failures
- Exhaust gas temperature sensor - Pre catalytic converter circuit short circuit to ground, short circuit to power, open circuit, high resistance
- Connector is disconnected, connector pin is backed out, connector pin corrosion
- Exhaust gas temperature sensor - Pre catalytic converter failure

diagnostic equipment check data
signals

- 0x03F7 Exhaust Gas Temperature Bank2 Sensor 1
- 0x03E7 Exhaust Gas Temperature Bank2 Sensor 1 Voltage
- Refer to the electrical circuit diagram and check the exhaust gas temperature sensor - Pre catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance
- Check and install a new exhaust gas temperature sensor - Pre catalytic converter as required
- Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

P208A-13	Reductant Pump "A" Control Circuit / Open - Circuit open	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid tank module - output from the ECM to DEF pump control for open circuit ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference RAPM ▪ Monitor Description ▪ Diesel exhaust fluid tank module - output from the ECM to DEF pump control open circuit ▪ Prioritised List of Possible Causes ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Diesel exhaust fluid tank module circuit open circuit, high resistance - output from the ECM to DEF pump control ▪ Connector is disconnected, connector pin is	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The electrical checks normally occur once ignition is on or after engine start ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit open circuit, high resistance - output from the ECM to DEF pump control ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ On completion of the previous repair actions the following MU be carried out ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest ▪ NOTE: The engine MIL is OFF (when engine ON) ▪ Start engine ▪ The 'inducement' warning clear automatically once the monitor has executed and passed with no malfunction. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle, then the
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		backed out, connector pin corrosion	malfunction has re-occurred. Further repair is required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab - NOTE: This clears the fault stor from P2BAE / P2BAF
P208B-00	Reductant Pump "A" Control Range/Performance - No sub type information	- Purpose of the DTC - To monitor diesel exhaust fluid line pressure build up for over pressure - Electrical Cause - No - Mechanical Cause - Yes - Control Module Cavity - Not applicable - Monitor Description - Diesel exhaust fluid line pressure build up failure - Prioritised List of Possible Causes - Diesel exhaust fluid line not connected - Diesel exhaust fluid line mechanical integrity - Diesel exhaust fluid injector mechanical integrity - Diesel exhaust fluid injector stuck open - Diesel exhaust fluid line heater frozen -	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The pressure build-up normally occurs once the selective catalyst reduction catalytic converter temperature exceeds 150 °C - Prioritised Checks to Perform - Check diesel exhaust fluid line correctly installed - Check diesel exhaust fluid line mechanical integrity - Check diesel exhaust fluid injector for mechanical integrity - Check diesel exhaust fluid injector is not stuck open - Check diesel exhaust fluid line heater is not frozen - restricted - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance - monitor for the pressure build up - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		restricted flow ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid injection pump failure	▪ Contact dealer technical support ▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF system pressurized ▪ On completion of the previous repair actions the following MU be carried out ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest ▪ NOTE: The engine MIL is OFF (when engine ON) ▪ Drive vehicle at urban speeds approximately 50mph ▪ NOTE: These conditions ensure that the selective catalyst reduction catalytic converter temperature exceeds 150 °C at which the sy pressure build up is initiated ▪ The monitoring is executed with 15 minutes of the driving cycle ▪ The 'inducement' warning clear automatically once the monitor has executed. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle: the malfunction has re-occurred ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This clears the fault stored from P2BAE / P2BAF
P208B-4B	Reductant Pump "A" Control Range/Performance - Over temperature	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid tank module - output from the ECM to DEF pump control for over temperature ▪ Electrical Cause	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module control circuit for short circuit to power output from the ECM to DEF pump ▪ Check for over heating of the

		▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference RAPM ▪ Monitor Description ▪ Diesel exhaust fluid tank module - output from the ECM to DEF pump control over temperature ▪ Prioritised List of Possible Causes ▪ The engine control module detected an internal temperature above the expected range ▪ Diesel exhaust fluid tank module has over heated ▪ Diesel exhaust fluid tank module circuit short circuit to power - output from the ECM to DEF pump control ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	module or evidence of over heat near to the module ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P208B-64	Reductant Pump "A" Control Range/Performance - Signal plausibility failure	▪ Purpose of the DTC ▪ To monitor for the pump pressure ▪ Electrical Cause	▪ Vehicle Conditions to enable DTC Logging strategy ▪ DEF system pressurized ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit

		▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid injection pump range performance ▪ Prioritised List of Possible Causes ▪ The engine control module detected plausibility failures ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid injection pump failure	diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance - monitor for the pur pressure build up ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P208C-00	Reductant Pump "A" Control Circuit Low - No sub type information	▪ Diesel exhaust fluid tank module circuit short circuit to ground ▪ Output from the ECM to DEF pump control ▪ Connector is disconnected, connector pin is backed out, connector pin	▪ Refer to the electrical circuit diagram and check the diesel exhaust fluid tank module circuit for short circuit to ground ▪ output from the ECM to DEF pump control ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the Jaguar Land Rover approved

		corrosion	diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P208C-11	Reductant Pump "A" Control Circuit Low - Circuit short to ground	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid tank module - output from the ECM to DEF pump control for short circuit to ground ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference RAPM ▪ Monitor Description ▪ Diesel exhaust fluid tank module control short circuit to ground - output from the ECM to DEF pump ▪ Prioritised List of Possible Causes ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Diesel exhaust fluid tank module circuit short circuit to	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground - output from the ECM to DEF pump ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		ground - output from the ECM to DEF pump Connector is disconnected, connector pin is backed out, connector pin corrosion	
P208D-12	Reductant Pump "A" Control Circuit High - Circuit short to battery	- Purpose of the DTC - To monitor diesel exhaust fluid tank module - output from the ECM to DEF pump control for short circuit to power - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference RAPM - Monitor Description - Diesel exhaust fluid tank module control short circuit to power - output from the ECM to DEF pump - Prioritised List of Possible Causes - The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another	- Vehicle Conditions to enable DTC Logging Strategy - Engine running - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to power - output from the ECM to DEF pump - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Contact dealer technical support - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		value was expected ▪ Diesel exhaust fluid tank module circuit short circuit to power - output from the ECM to DEF pump ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	
P208E-73	Reductant Injection Valve Stuck Closed (Bank 1 Unit 1) - Actuator stuck closed	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid injector opening / closing ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid injector is stuck closed ▪ Prioritised List of Possible Causes ▪ The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to open some piece of equipment ▪ Diesel exhaust fluid injector	▪ Vehicle Conditions to enable DTC Logging strategy ▪ DEF system pressurized ▪ Prioritised Checks to Perform ▪ Check and install a new diesel exhaust fluid injector as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		failure	
P209F-16	Reductant Tank Heater Control Performance - Circuit voltage below threshold	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid heater control unit ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid heater control unit voltage too low ▪ Prioritised List of Possible Causes ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Diesel exhaust fluid heater control unit circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	▪ Vehicle Conditions to enable DTC Logging strategy ▪ DEF heater on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit if required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

P209F-17	Reductant Tank Heater Control Performance - Circuit voltage above threshold	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid heater control unit ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid heater control unit voltage too high ▪ Prioritised List of Possible Causes ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Diesel exhaust fluid heater control unit circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	▪ Vehicle Conditions to enable DTC Logging strategy ▪ DEF heater on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit if required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

P209F-18	Reductant Tank Heater Control Performance - Circuit current below threshold	■ Purpose of the DTC ■ To monitor diesel exhaust fluid heater control unit ■ Electrical Cause ■ Yes ■ Mechanical Cause ■ No ■ Control Module Cavity ■ Not applicable ■ Monitor Description ■ Diesel exhaust fluid heater control unit current too low ■ Prioritised List of Possible Causes ■ The engine control module has measured current flow below a specified range ■ Diesel exhaust fluid heater control unit circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Diesel exhaust fluid heater control unit failure	■ Vehicle Conditions to enable DTC Logging strategy ■ DEF heater on ■ Prioritised Checks to Perform ■ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Check and install a new diesel exhaust fluid heater control unit if required ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P209F-19	Reductant Tank Heater Control Performance - Circuit current above	■ Purpose of the DTC ■ To monitor diesel exhaust	■ Vehicle Conditions to enable DTC Logging strategy ■ DEF heater on

		threshold	fluid heater control unit - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Not applicable - Monitor Description - Diesel exhaust fluid heater control unit current too high - Prioritised List of Possible Causes - The engine control module has measured current flow above a specified range - Diesel exhaust fluid heater control unit circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Diesel exhaust fluid heater control unit failure	- Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new diesel exhaust fluid heater control unit if required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P209F-96	Reductant Tank Heater Control Circuit Performance - Component internal failure			- Check diesel exhaust fluid line heater for mechanical integrity - Check diesel exhaust fluid line heater for correct installation - Refer to the electrical circuit diagram

		NOTE: To monitor diesel exhaust fluid line heater ▪ Diesel exhaust fluid line heater mechanical integrity ▪ Diesel exhaust fluid line heater installation ▪ Diesel exhaust fluid line heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid line heater failure	and check the diesel exhaust fluid to module circuit for open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P20B9-13	Reductant Heater A Control Circuit / Open - Circuit open	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid line heater ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid line heater open circuit, high resistance ▪ Prioritised List of	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid line heater circuit open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid line heater as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		Possible Causes ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Diesel exhaust fluid line heater circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid line heater failure	
P20BA-16	Reductant Heater A Control Performance - Circuit voltage below threshold	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid line heater ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid line heater voltage too low ▪ Prioritised List of Possible Causes ▪ The engine control module measured a voltage below a	▪ Vehicle Conditions to enable DTC Logging strategy ▪ DEF heater on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid line heater circuit open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid line heater as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		specified range but not necessarily a short circuit to ground ▪ Diesel exhaust fluid line heater circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid line heater failure	
P20BA-17	Reductant Heater A Control Performance - Circuit voltage above threshold	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid line heater ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid line heater voltage too high ▪ Prioritised List of Possible Causes ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Diesel exhaust fluid line heater	▪ Vehicle Conditions to enable DTC Logging strategy ▪ DEF heater on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid line heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid line heater as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid line heater failure	
P20BA-18	Reductant Heater A Control Performance - Circuit current below threshold	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid line heater ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid line heater current too low ▪ Prioritised List of Possible Causes ▪ The engine control module has measured current flow below a specified range ▪ Diesel exhaust fluid line heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is	▪ Vehicle Conditions to enable DTC Logging strategy ▪ DEF heater on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid line heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid line heater as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid line heater failure	
P20BA-19	Reductant Heater A Control Performance - Circuit current above threshold	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid line heater ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid line heater current too high ▪ Prioritised List of Possible Causes ▪ The engine control module has measured current flow above a specified range ▪ Diesel exhaust fluid line heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid line heater	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF heater on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid line heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid line heater as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		failure	
P20BB-11	Reductant Heater A Control Circuit Low - Circuit short to ground	■ Purpose of the DTC ■ To monitor diesel exhaust fluid line heater ■ Electrical Cause ■ Yes ■ Mechanical Cause ■ No ■ Control Module Cavity ■ Not applicable ■ Monitor Description ■ Diesel exhaust fluid line heater short circuit to ground ■ Prioritised List of Possible Causes ■ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ■ Diesel exhaust fluid line heater circuit short circuit to ground ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Diesel exhaust fluid line heater failure	■ Vehicle Conditions to enable DTC Logging Strategy ■ Engine running ■ Prioritised Checks to Perform ■ Refer to the electrical circuit diagrams and check the diesel exhaust fluid line heater circuit short circuit to ground ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Check and install a new diesel exhaust fluid line heater as required ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

P20BC-12	Reductant Heater A Control Circuit High - Circuit short to battery	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid line heater ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid line heater short circuit to power ▪ Prioritised List of Possible Causes ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Diesel exhaust fluid line heater circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid line heater failure	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid line heater circuit short circuit to power ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid line heater as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P20E8-84	Reductant Pressure Too Low - Signal below	▪ Purpose of the DTC ▪ To monitor for	

allowable range

diesel exhaust
fluid line under
pressure error

- Electrical Cause
 - No
- Mechanical Cause
 - Yes
- Control Module Cavity
 - Not applicable
- Monitor Description
 - If modelled pressure falls below 2 bar system is purged and pressure is built up again. If this happens 12 times during driving cycle error is set (Fast path due to air bubbles)
- Prioritised List of Possible Causes
 - The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range
 - DEF injector dosing too high. Diesel Exhaust Fluid injection pump delivers too much DEF
 - Diesel exhaust fluid line leakage
 - Diesel exhaust fluid line air bubbles
 - Diesel exhaust fluid tank module circuit

NOTE:

A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The pressure build-up normally occurs once the selective catalyst reduction catalytic converter temperature exceeds 150 °C

- Prioritised Checks to Perform
 - Check diesel exhaust fluid line leakage
 - Using the Jaguar Land Rover approved diagnostic equipment carry out diagnostic routine 0x4
 - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance
 - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
 - Contact dealer technical support
- Vehicle Conditions to enable DTC Logging Strategy
 - DEF system pressurized
 - On completion of the previous repair actions the following MU be carried out
 - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
 - NOTE: The engine MIL is OFF (when engine ON)
 - Drive vehicle at urban speeds approximately 50mph
 - NOTE: These conditions ensure

		short circuit to ground, short circuit to power, open circuit, high resistance Connector is disconnected, connector pin is backed out, connector pin corrosion	that the selective catalyst reduction catalytic converter temperature exceeds 150 °C at which the sy pressure build up is initiated - The monitoring is executed with 15 minutes of the driving cycle - The 'inducement' warning clear automatically once the monitoring has executed. This terminates the drive cycle - NOTE: If the engine MIL shows during the driving cycle: the malfunction has re-occurred - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab - NOTE: This clears the fault stored from P2BAE / P2BAF
P20E8-85	Reductant Pressure Too Low - Signal above allowable range	- Purpose of the DTC - To monitor for diesel exhaust fluid line under pressure error - Electrical Cause - No - Mechanical Cause - Yes - Control Module Cavity - Not applicable - Monitor Description - If modelled pressure falls below 4.5 bar the system is purged and pressure is built up again. If this occurs during a drive cycle the error is set - Prioritised List of Possible Causes - The engine control module	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The pressure build-up normally occurs once the selective catalyst reduction catalytic converter temperature exceeds 150 °C - Prioritised Checks to Perform - Check diesel exhaust fluid line for leakage - Using the Jaguar Land Rover approved diagnostic equipment carry out diagnostic routine 0x4 - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high

		has determined failures where some circuit quantity, reported via serial data, is above a specified range ▪ DEF injector dosing too high. Diesel Exhaust Fluid injection pump delivers too much DEF ▪ Diesel exhaust fluid line leakage ▪ Diesel exhaust fluid line air bubbles ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF system pressurized ▪ On completion of the previous repair actions the following MU be carried out ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest ▪ NOTE: The engine MIL is OFF (when engine ON) ▪ Drive vehicle at urban speeds approximately 50mph ▪ NOTE: These conditions ensure that the selective catalyst reduction catalytic converter temperature exceeds 150 °C at which the sy pressure build up is initiated ▪ The monitoring is executed with 15 minutes of the driving cycle ▪ The 'inducement' warning clear automatically once the monitor has executed. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle: the malfunction has re-occurred ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This clears the fault stored from P2BAE / P2BAF
P20E9-84	Reductant Pressure Too High - Signal below allowable range	▪ Purpose of the DTC ▪ To monitor for over pressure (slow) ▪ Electrical Cause	

				▪ No	
				▪ Mechanical Cause ▪ Yes	
				▪ Control Module Cavity ▪ Not applicable	
				▪ Monitor Description ▪ If modelled pressure rises above 8,5 bar system is purged and pressure is built up again. If this happens 5 times during driving cycle error is set	
				▪ Prioritised List of Possible Causes ▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ▪ Diesel exhaust fluid line frozen / blocked ▪ Diesel exhaust fluid line heater frozen ▪ Diesel exhaust fluid injector frozen / blocked ▪ Diesel exhaust fluid tank module frozen	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The pressure build-up normally occurs once the selective catalyst reduction catalytic converter temperature exceeds 150 °C
					▪ Prioritised Checks to Perform ▪ Check diesel exhaust fluid line for blockage ▪ Check diesel exhaust fluid injector for blockage ▪ Check diesel exhaust fluid tank module for blockage ▪ Check diesel exhaust fluid line is not frozen ▪ Check diesel exhaust fluid injector is not frozen ▪ Check diesel exhaust fluid tank module is not frozen ▪ Vehicle Conditions to enable DTC Logging strategy ▪ DEF system pressurized ▪ On completion of the previous repair actions the following MU be carried out ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest ▪ NOTE: The engine MIL is OFF (when engine ON) ▪ Drive vehicle at urban speeds approximately 50mph ▪ NOTE: These conditions ensure that the selective catalyst reduction catalytic converter temperature

			exceeds 150 °C at which the sy pressure build up is initiated ▪ The monitoring is executed wit 15 minutes of the driving cycle ▪ The 'inducement' warning clear automatically once the monitor has executed. This terminates t drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle: the malfunction has re-occurred ▪ Using the Jaguar Land Rover approved diagnostic equipmen clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This cleans up the fault storage from P2BAE/P2BAF
P20E9-85	Reductant Pressure Too High - Signal above allowable range	▪ Purpose of the DTC ▪ To monitor for over pressure (fast) ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ If modelled pressure rises above 10 bar, system is purged and pressure is built up again. If this happens 5 times during driving cycle error is set ▪ Prioritised List of Possible Causes ▪ The engine control module has determined failures where some circuit	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The pressure build-up normally occurs once the selective catalyst reduction catalytic converter temperature exceeds 150 °C ▪ Prioritised Checks to Perform ▪ Check diesel exhaust fluid line blockage ▪ Check diesel exhaust fluid injected for blockage ▪ Check diesel exhaust fluid tank module for blockage ▪ Check diesel exhaust fluid line not frozen ▪ Check diesel exhaust fluid injected is not frozen ▪ Check diesel exhaust fluid tank

		quantity, reported via serial data, is above a specified range ▪ Diesel exhaust fluid line frozen / blocked ▪ Diesel exhaust fluid line heater frozen ▪ Diesel exhaust fluid injector frozen / blocked ▪ Diesel exhaust fluid tank module frozen	module is not frozen ▪ Vehicle Conditions to enable DTC Logging strategy ▪ DEF system pressurized ▪ On completion of the previous repair actions the following MU be carried out ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest ▪ NOTE: The engine MIL is OFF (when engine ON) ▪ Drive vehicle at urban speeds approximately 50mph ▪ NOTE: These conditions ensure that the selective catalyst reduction catalytic converter temperature exceeds 150 °C at which the sy pressure build up is initiated ▪ The monitoring is executed with 15 minutes of the driving cycle ▪ The 'inducement' warning clear automatically once the monitor has executed. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle: the malfunction has re-occurred ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This cleans up the fault storage from P2BAE/P2BAF
P20EE-00	SCR NOx Catalyst Efficiency Below Threshold (Bank 1) - No sub type information	▪ Purpose of the DTC ▪ Monitors SCR Nox catalyst efficiency - reduced catalyst Nox efficiency ▪ The SCR device ammonia storage capability is	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Drive vehicle at urban speeds until SCR catalyst warm and exhaust sensors active. Ensure the vehicle not in DPF active regeneration drive at urban speeds for 20min / 60mph) ▪ Prioritised Checks to Perform ▪ Damaged/Removed SCR catalyst

		greatly reduced or the device is missing - Electrical Cause - No - Mechanical Cause - Yes - Control Module Cavity - Not applicable - Monitor Description - SCR - reduced catalyst Nox conversion efficiency detected by passive monitoring but not detected by active monitoring; on two consecutive occasions - The SCR device ammonia storage capability is greatly reduced or the device is missing - Prioritised List of Possible Causes - Defective SCR catalyst (aged) - Deficient DEF reagent delivery - Diesel exhaust fluid injector partial delivery - Damaged exhaust metal work - SCR catalyst damage	- Deposits on front face of SCR catalyst - Check diesel exhaust fluid injector for deposits or corrosion - Check exhaust metal work for damage - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P20EE-04	SCR NOx Catalyst Efficiency Below Threshold	- Purpose of the DTC - Selective	Vehicle Conditions to enable DTC Logging strategy

(Bank 1) - System internal failures

catalytic reduction - reduced catalyst Nox conversion efficiency by passive monitoring

- The SCR device ammonia storage capability is greatly reduced or the device is missing
- Electrical Cause
 - No
- Mechanical Cause
 - Yes
- Control Module Cavity
 - Not applicable
- Monitor Description
 - SCR - reduced catalyst Nox conversion efficiency detected by passive monitoring but not detected by active monitoring; on two consecutive occasions
 - The SCR device ammonia storage capability is greatly reduced or the device is missing
- Prioritised List of Possible Causes
 - Defective SCR catalyst (aged)
 - Deficient DEF reagent delivery
 - Diesel exhaust fluid injector

- Drive vehicle at urban speeds until SCR catalyst warm and exhaust sensors active. Ensure the vehicle is not in DPF active regeneration (drive at urban speeds for 20min / 60mph)
- Prioritised Checks to Perform
 - Damaged/Removed SCR catalyst
 - Deposits on front face of SCR catalyst
 - Check diesel exhaust fluid injector for deposits or corrosion
 - Check exhaust metal work for damage
 - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		partial delivery ▪ Damaged exhaust metal work ▪ SCR catalyst damage	
P20EE-92	SCR NOx Catalyst Efficiency Below Threshold (Bank 1) - Performance or incorrect operation	▪ Purpose of the DTC ▪ Selective catalytic reduction - reduced catalyst Nox conversion efficiency in ammonia storage mode ▪ The SCR device ammonia storage capability is greatly reduced or the device is missing ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ SCR - reduced catalyst Nox conversion efficiency detected by passive monitoring but not detected by active monitoring; on two consecutive occasions ▪ The SCR device ammonia storage capability is greatly reduced	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Drive vehicle at urban speeds until SCR catalyst warm and exhaust sensors active. Ensure the vehicle not in DPF active regeneration drive at urban speeds for 20min / 60mph) ▪ Prioritised Checks to Perform ▪ Under hood checks ▪ Check both spring band clips on crankcase breather (rubber hoses are correctly positioned and in 'clamped' condition (grenade clip pulled to allow full clamping pressure) ▪ Check upper spring band clip on bellows joint to low pressure EGR cooler is fitted correctly and is not rolled in-ward upon itself. Ensure no damage is evident in anyway. Ensure grenade clip is pulled and the clip is in its clamped position ▪ Check low pressure EGR cooler 'clam' clamp is correctly seated evenly around its circumference, it 'bridges' either side of the manifold low pressure EGR and exhaust flanges. Visually check the clam once tightened to the correct torque and ensure no thread form is visible outside of the 'boxed' area. If thread form is visible then this indicates a mis-build condition and a re-work is required ▪ Under vehicle checks ▪ Damaged/Removed SCR catalyst ▪ Deposits on front face of SCR catalyst ▪ Check diesel exhaust fluid injection

		- or the device is missing ■ Prioritised List of Possible Causes ■ Low pressure exhaust gas recirculation bellows incorrect installation between low pressure EGR and low pressure EGR mixer duct ■ Low pressure exhaust gas recirculation cooler clam clamp leakage or incorrect installation ■ Defective SCR catalyst (aged) ■ Deficient DEF reagent delivery ■ Diesel exhaust fluid injector partial delivery ■ Damaged exhaust metal work ■ SCR catalyst damage	for deposits or corrosion ■ Check exhaust metal work for damage ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P20EE-96	SCR NOx Catalyst Efficiency Below Threshold (Bank 1) - Component internal failure	■ Purpose of the DTC ■ Selective catalytic reduction - reduced catalyst Nox conversion efficiency by active monitoring ■ The SCR device ammonia storage capability is greatly reduced or the device is	■ Vehicle Conditions to enable DTC Logging strategy ■ Drive vehicle at urban speeds until SCR catalyst warm and exhaust sensors active. Ensure the vehicle not in DPF active regeneration drive at urban speeds for 20min / 60mph) ■ Prioritised Checks to Perform ■ Damaged/Removed SCR catalyst ■ Deposits on front face of SCR catalyst ■ Check diesel exhaust fluid injector for deposits or corrosion

		missing - Electrical Cause - No - Mechanical Cause - Yes - Control Module Cavity - Not applicable - Monitor Description - SCR - reduced catalyst Nox conversion efficiency detected by active monitoring but not detected by active monitoring; on two consecutive occasions - The SCR device ammonia storage capability is greatly reduced or the device is missing - Prioritised List of Possible Causes - Defective SCR catalyst (aged) - Deficient DEF reagent delivery - Diesel exhaust fluid injector partial delivery - Damaged exhaust metal work - SCR catalyst damage	- Check exhaust metal work for damage - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P20EE-97	SCR NOx Catalyst Efficiency Below Threshold (Bank 1) - Component or System operation	- Purpose of the DTC - Selective catalytic reduction -	- Vehicle Conditions to enable DTC Logging strategy - Drive vehicle at urban speeds u SCR catalyst warm and exhaust

obstructed or blocked	reduced catalyst Nox conversion efficiency in ammonia pressure control mode ▪ The SCR device ammonia storage capability is greatly reduced or the device is missing ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ SCR - reduced catalyst Nox conversion efficiency detected by passive monitoring but not detected by active monitoring; on two consecutive occasions ▪ The SCR device ammonia storage capability is greatly reduced or the device is missing ▪ Prioritised List of Possible Causes ▪ Defective SCR catalyst (aged) ▪ Deficient DEF reagent delivery ▪ Diesel exhaust fluid injector partial delivery	sensors active. Ensure the vehicle is not in DPF active regeneration drive at urban speeds for 30min (20 mph), with light throttle pedal transients ▪ Prioritised Checks to Perform ▪ Damaged/Removed SCR catalyst ▪ Deposits on front face of SCR catalyst ▪ Check diesel exhaust fluid injector for deposits or corrosion ▪ Check exhaust metal work for damage ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
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		▪ Damaged exhaust metal work ▪ SCR catalyst damage	
P20FA-13	Reductant Pump "B" Control Circuit / Open - Circuit open	▪ Purpose of the DTC ▪ To monitor DEF back flow pump for open circuit ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference RAPP ▪ Monitor Description ▪ Diesel exhaust fluid tank module - DEF back flow pump circuit open circuit, high resistance ▪ Prioritised List of Possible Causes ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Diesel exhaust fluid tank module circuit open circuit, high resistance - DEF back flow pump ▪ Connector is disconnected,	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit open circuit, high resistance - DEF back flow pump ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		connector pin is backed out, connector pin corrosion	
P20FB-00	Reductant Pump "B" Control Range/Performance - No sub type information	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid tank module - DEF back flow pump ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid tank module - DEF back flow pump system pressure can not be reduced after ignition off ▪ Prioritised List of Possible Causes ▪ Diesel exhaust fluid tank module - DEF back flow pump circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Vehicle Conditions to enable DTC Logging strategy ▪ DEF system pressurized ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance - DEF back flow pump ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P20FC-11	Reductant Pump "B" Control Circuit Low -	▪ Purpose of the DTC ▪ To monitor DEF	▪ Vehicle Conditions to enable DTC Logging strategy

		Circuit short to ground back flow pump for short circuit to ground - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference RAPP - Monitor Description - Diesel exhaust fluid tank module - DEF back flow pump circuit short circuit to ground - Prioritised List of Possible Causes - The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Diesel exhaust fluid tank module circuit short circuit to ground - DEF back flow pump - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Engine running - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground - DEF back flow pump - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Contact dealer technical support - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P20FD-12	Reductant Pump "B" Control Circuit High -	- Purpose of the DTC - To monitor DEF	Vehicle Conditions to enable DTC Logging strategy

		Circuit short to battery back flow pump for short circuit to power - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference RAPP - Monitor Description - Diesel exhaust fluid tank module - DEF back flow pump circuit short circuit to power - Prioritised List of Possible Causes - The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Diesel exhaust fluid tank module circuit short circuit to power - DEF back flow pump - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Engine running - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to power - DEF back flow pump - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Contact dealer technical support - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P20FE-00	Reductant Metering Unit Performance - No sub type	- Purpose of the DTC - Monitors DEF	Vehicle Conditions to enable DTC Logging strategy

	information	reagent delivery performance to the SCR device - Electrical Cause - No - Mechanical Cause - Yes - Control Module Cavity - Not applicable - Monitor Description - An active overdosing of the SCR is performed following a passive monitoring event (lack of apparent SCR efficiency). The reagent delivery provided by the dosing valve is monitored during this process - Prioritised List of Possible Causes - Defective DEF reagent delivery - Defective SCR catalyst (aged) - Deficient DEF reagent delivery - Diesel exhaust fluid injector partial delivery - Damaged exhaust metal work - SCR catalyst damage	- Drive vehicle at urban speeds until SCR catalyst warm and exhaust sensors active. Ensure the vehicle not in DPF active regeneration (drive at urban speeds for 20min / 60mph) - Prioritised Checks to Perform - Damaged/Removed dosing module - Dosing module stuck closed / seized by deposits - Deposits on SCR device front face - Damaged/Removed SCR catalyst - Deposits on front face of SCR catalyst - Check diesel exhaust fluid injector for deposits or corrosion - Check exhaust metal work for damage
P20FE-97	Reductant Metering Unit Performance - Component or system operation	- Purpose of the DTC - Selective catalytic	- Vehicle Conditions to enable DTC Logging strategy - Drive vehicle at urban speeds until

		obstructed or blocked reduction - diesel exhaust fluid delivery efficiency by active monitoring - Electrical Cause - No - Mechanical Cause - Yes - Control Module Cavity - Not applicable - Monitor Description - Selective catalytic reduction - Nox conversion efficiency is below threshold; the diesel exhaust fluid dosing module delivery is greatly reduced - Prioritised List of Possible Causes - Defective DEF reagent delivery - Defective SCR catalyst (aged) - Deficient DEF reagent delivery - Diesel exhaust fluid injector partial delivery - Damaged exhaust metal work - SCR catalyst damage	SCR catalyst warm and exhaust sensors active. Ensure the vehicle is not in DPF active regeneration (drive at urban speeds for 20min / 60mph) - Prioritised Checks to Perform - Damaged/Removed dosing module - Dosing module stuck closed / stuck open by deposits - Deposits on SCR device front face - Damaged/Removed SCR catalyst - Deposits on front face of SCR catalyst - Check diesel exhaust fluid injector for deposits or corrosion - Check exhaust metal work for damage
P2100-13	Throttle Actuator A Control Motor Circuit /Open - Circuit open	The engine control module has determined an open circuit via lack of bias voltage, low current	- Diagnosis of this DTC may require the use of the Jaguar Land Rover approved diagnostic equipment check data link signals 0x040B Throttle Motor PWM Signal

		flow, no change in the state of an input in response to an output ▪ Electric throttle motor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Electric throttle failure	- Commanded ▪ 0x0917 Throttle Position Sensor Voltage - Sensor 1 ▪ 0xF411 Absolute Throttle Position ▪ 0x0530 Absolute Throttle Position Bank 1 ▪ Refer to the electrical circuit diagram and check the electric throttle motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new electric throttle as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out 'Air path set-up routine's (0x4030/0x4051_01/0x4050_01/0xF405) ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out 'Engine control module low pressure EGR adaption clear routine (0x0406_0C) ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P2101-74	Throttle Actuator "A" Control Motor Circuit Range/Performance - Actuator slipping	▪ Electric throttle valve obstructed ▪ Electric throttle valve contaminated ▪ Electric throttle valve mechanical integrity ▪ Electric throttle motor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected,	▪ Check electric throttle valve is not obstructed and valve spring is operating correctly ▪ Check electric throttle valve for mechanical integrity and contamination ▪ Diagnosis of this DTC may require use the Jaguar Land Rover approved diagnostic equipment check data log signals ▪ 0x040B Throttle Motor PWM Signal - Commanded ▪ 0x0917 Throttle Position Sensor Voltage - Sensor 1 ▪ 0xF411 Absolute Throttle Position

		connector pin is backed out, connector pin corrosion Electric throttle failure	0x0530 Absolute Throttle Position Bank 1 - Refer to the electrical circuit diagram and check the electric throttle motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new electric throttle as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Air path set-up routine's (0x4030/0x4051_01/0x4050_01/0xF405) - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Engine control module low pressure EGR adaption clear routine (0x0406_0C) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P2101-77	Throttle Actuator A Control Motor Circuit Range/Performance - Commanded position not reachable	- The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment - Electric throttle valve obstructed - Electric throttle valve contaminated - Electric throttle valve mechanical integrity - Electric throttle motor circuit short	- Check electric throttle valve is not obstructed and valve spring is operating correctly - Check electric throttle valve for mechanical integrity and contamination - Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data logger signals 0x040B Throttle Motor PWM Signal - Commanded 0x0917 Throttle Position Sensor Voltage - Sensor 1 0xF411 Absolute Throttle Position 0x0530 Absolute Throttle Position Bank 1 - Refer to the electrical circuit diagram and check the electric throttle motor circuit for short circuit to ground, short

		circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Electric throttle failure	circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new electric throttle cable as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Air path set-up routine' (0x4030/0x4051_01/0x4050_01/0xF405) - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Engine control module low pressure EGR adaption clear routine' (0x0406_0C) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P2102-11	Throttle Actuator A Control Motor Circuit Low - Circuit short to ground	- The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Electric throttle motor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Electric throttle failure	- Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data log signals 0x040B Throttle Motor PWM Signal - Commanded 0x0917 Throttle Position Sensor Voltage - Sensor 1 0xF411 Absolute Throttle Position 0x0530 Absolute Throttle Position Bank 1 - Refer to the electrical circuit diagram and check the electric throttle motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new electric throttle cable as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Air path set-up routine'

			routine's (0x4030/0x4051_01/0x4050_01 0xF405) - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Engine control module low pressure EGR adaption clear routine (0x0406_0C) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P2103-12	Throttle Actuator A Control Motor Circuit High - Circuit short to battery	- The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Electric throttle motor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Electric throttle failure	- Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data list for signals 0x040B Throttle Motor PWM Signal - Commanded 0x0917 Throttle Position Sensor Voltage - Sensor 1 0xF411 Absolute Throttle Position 0x0530 Absolute Throttle Position Bank 1 - Refer to the electrical circuit diagram and check the electric throttle motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new electric throttle as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Air path set-up routine's (0x4030/0x4051_01/0x4050_01 0xF405) - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Engine control module low pressure EGR adaption clear routine (0x0406_0C) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored

			DTCs using the Diagnosis Menu take and retest
P2122-00	Throttle/Pedal Position Sensor/Switch D Circuit Low - No sub type information	- Accelerator pedal position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Accelerator pedal position sensor failure	- Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data for pedal position sensor signals 0x0914 Pedal Position Sensor Voltage - Sensor 1 - Volts - Value should rise and fall with accelerator pedal application. (~700mV - 4800mV) - Refer to the electrical circuit diagram and check the accelerator pedal position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new accelerator pedal position sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take and retest
P2123-00	Throttle/Pedal Position Sensor/Switch D Circuit High - No sub type information	- Accelerator pedal position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Accelerator pedal position sensor failure	- Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data for pedal position sensor signals 0x0914 Pedal Position Sensor Voltage - Sensor 1 - Volts - Value should rise and fall with accelerator pedal application. (~700mV - 4800mV) - Refer to the electrical circuit diagram and check the accelerator pedal position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new accelerator pedal position sensor as required

			Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2127-00	Throttle/Pedal Position Sensor/Switch E Circuit Low - No sub type information	- Accelerator pedal position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Accelerator pedal position sensor failure	- Diagnosis of this DTC may require the Jaguar Land Rover approved diagnostic equipment check data list signals 0x0915 Pedal Position Sensor Voltage - Sensor 2 - Volts - Value should rise and fall with accelerator pedal application. (~350mV - 2500mV) - Refer to the electrical circuit diagram and check the accelerator pedal position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new accelerator pedal position sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2128-00	Throttle/Pedal Position Sensor/Switch E Circuit High - No sub type information	- Accelerator pedal position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Accelerator pedal position sensor failure	- Diagnosis of this DTC may require the Jaguar Land Rover approved diagnostic equipment check data list signals 0x0915 Pedal Position Sensor Voltage - Sensor 2 - Volts - Value should rise and fall with accelerator pedal application. (~350mV - 2500mV) - Refer to the electrical circuit diagram and check the accelerator pedal position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new accelerator pedal position sensor as required

			pedal position sensor as required Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take the vehicle for a test drive and retest
P2138-00	Throttle/Pedal Position Sensor/Switch D / E Voltage Correlation - No sub type information	- Accelerator pedal position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Accelerator pedal position sensor failure	- Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data for pedal position signals 0x0914 Pedal Position Sensor Voltage - Sensor 1 - Volts - Value should rise and fall with accelerator pedal application. (~700mV - 4800mV) 0x0915 Pedal Position Sensor Voltage - Sensor 2 - Volts - Value should rise and fall with accelerator pedal application. (~350mV - 2500mV) - Refer to the electrical circuit diagram and check the accelerator pedal position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new accelerator pedal position sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take the vehicle for a test drive and retest
P213A-13	Exhaust Gas Recirculation Throttle Control Circuit B / Open - Circuit open	- The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Exhaust gas recirculation - Low pressure butterfly	- Refer to the electrical circuit diagram and check the exhaust gas recirculation - Low pressure butterfly valve circuit for open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check exhaust gas recirculation - Low pressure butterfly valve for mechanical integrity - Linkage not seized

		- valve circuit open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation - Low pressure butterfly valve mechanical integrity - Exhaust gas recirculation - Low pressure butterfly valve failure	- Butterfly valve not stuck - Check and install a new exhaust gas recirculation - Low pressure butterfly valve as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Exhaust flap adaptation routine (0x0406_0D) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take and retest
P213B-72	Exhaust Gas Recirculation Throttle Control Circuit B Range/Performance - Actuator stuck open	- The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to close some piece of equipment - Exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation - Low pressure butterfly valve mechanical integrity - Exhaust gas recirculation - Low	- Refer to the electrical circuit diagram and check the exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check exhaust gas recirculation - Low pressure butterfly valve for mechanical integrity - Linkage not seized - Butterfly valve not stuck - Check and install a new exhaust gas recirculation - Low pressure butterfly valve as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Exhaust flap adaptation routine (0x0406_0D) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take and retest

		pressure butterfly valve failure	
P213B-73	Exhaust Gas Recirculation Throttle Control Circuit B Range/Performance - Actuator stuck closed	▪ The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to open some piece of equipment ▪ Exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas recirculation - Low pressure butterfly valve mechanical integrity ▪ Exhaust gas recirculation - Low pressure butterfly valve failure	▪ Refer to the electrical circuit diagram and check the exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check exhaust gas recirculation - Low pressure butterfly valve for mechanical integrity ▪ Linkage not seized ▪ Butterfly valve not stuck ▪ Check and install a new exhaust gas recirculation - Low pressure butterfly valve as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out 'Exhaust flap adaptation' routine (0x0406_0D) ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P213B-77	Exhaust Gas Recirculation Throttle Control Circuit B Range/Performance - Commanded position not reachable	▪ The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment ▪ Exhaust gas	▪ Refer to the electrical circuit diagram and check the exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check exhaust gas recirculation - Low pressure butterfly valve for mechanical integrity ▪ Linkage not seized

		- recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation - Low pressure butterfly valve mechanical integrity - Exhaust gas recirculation - Low pressure butterfly valve failure	- Butterfly valve not stuck - Check and install a new exhaust gas recirculation - Low pressure butterfly valve as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Exhaust flap adaptation' routine (0x0406_0D) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P213C-11	Exhaust Gas Recirculation Throttle Control Circuit B Low - Circuit short to ground	- The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation - Low	- Refer to the electrical circuit diagram and check the exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check exhaust gas recirculation - Low pressure butterfly valve for mechanical integrity - Linkage not seized - Butterfly valve not stuck - Check and install a new exhaust gas recirculation - Low pressure butterfly valve as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Exhaust flap adaptation' routine (0x0406_0D) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		pressure butterfly valve mechanical integrity Exhaust gas recirculation - Low pressure butterfly valve failure	
P213C-16	Exhaust Gas Recirculation Throttle Control Circuit B Low - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation - Low pressure butterfly valve mechanical integrity - Exhaust gas recirculation - Low pressure butterfly valve failure	- Refer to the electrical circuit diagram and check the exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check exhaust gas recirculation - Low pressure butterfly valve for mechanical integrity - Linkage not seized - Butterfly valve not stuck - Check and install a new exhaust gas recirculation - Low pressure butterfly valve as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Exhaust flap adaptation routine (0x0406_0D) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take test and retest
P213D-12	Exhaust Gas Recirculation Throttle Control Circuit B High - Circuit short to battery	The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected	- Refer to the electrical circuit diagram and check the exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check exhaust gas recirculation - Low pressure butterfly valve for mechanical integrity

		- Exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation - Low pressure butterfly valve mechanical integrity - Exhaust gas recirculation - Low pressure butterfly valve failure	pressure butterfly valve for mechanical integrity - Linkage not seized - Butterfly valve not stuck - Check and install a new exhaust gas recirculation - Low pressure butterfly valve as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Exhaust flap adaptation' routine (0x0406_0D) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P213D-17	Exhaust Gas Recirculation Throttle Control Circuit B High - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation - Low pressure butterfly valve mechanical	- Refer to the electrical circuit diagram and check the exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check exhaust gas recirculation - Low pressure butterfly valve for mechanical integrity - Linkage not seized - Butterfly valve not stuck - Check and install a new exhaust gas recirculation - Low pressure butterfly valve as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Exhaust flap adaptation' routine (0x0406_0D) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest

		integrity Exhaust gas recirculation - Low pressure butterfly valve failure	
P213E-00	Fuel Injection System Fault - Forced Engine Shutdown - No sub type information	- Other fuel injection related DTCs - Engine control module failure	- Check engine control module for fuel injection related DTCs and refer to relevant DTC index - Check and install a new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'New Module - Power control module - Diesel' routine - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Fuel lift pump' routine (0x4027_00) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P2141-00	Exhaust Gas Recirculation Throttle Control Circuit A Low - No sub type information	- Electric throttle motor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Electric throttle failure	- Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data link signals 0x040B Throttle Motor PWM Signal - Commanded 0x0917 Throttle Position Sensor Voltage - Sensor 1 0xF411 Absolute Throttle Position 0x0530 Absolute Throttle Position Bank 1 - Refer to the electrical circuit diagram and check the electric throttle motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new electric throttle as required - Using the Jaguar Land Rover

			approved diagnostic equipment carry out 'Air path set-up routine's (0x4030/0x4051_01/0x4050_01/0xF405) - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Engine control module low pressure EGR adaption clear routine (0x0406_0C) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P2142-00	Exhaust Gas Recirculation Throttle Control Circuit A High - No sub type information	- Electric throttle motor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Electric throttle failure	- Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data list for signals 0x040B Throttle Motor PWM Signal - Commanded 0x0917 Throttle Position Sensor Voltage - Sensor 1 0xF411 Absolute Throttle Position 0x0530 Absolute Throttle Position Bank 1 - Refer to the electrical circuit diagram and check the electric throttle motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new electric throttle as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Air path set-up routine's (0x4030/0x4051_01/0x4050_01/0xF405) - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Engine control module low pressure EGR adaption clear routine (0x0406_0C)

			Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P214E-00	Reductant Pump "A" Current Too High - No sub type information	- The engine control module detected an internal temperature above the expected range - Diesel exhaust fluid tank module circuit short circuit to power - Output from the ECM to DEF pump control - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Refer to the electrical circuit diagram and check the diesel exhaust fluid tank module circuit for short circuit to power - output from the ECM to DEF pump control - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Contact dealer technical support - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P214E-19	Reductant Pump "A" Current Too High - Circuit current above threshold	- Purpose of the DTC - To monitor diesel exhaust fluid tank module - Supply pump - Electrical Cause - No - Mechanical Cause - Yes - Control Module Cavity - Not applicable - Monitor Description - Diesel exhaust fluid tank module - Supply pump coil resistance is not plausible - Prioritised List of Possible Causes - The engine control module	- Vehicle Conditions to enable DTC Logging strategy - DEF system pressurized - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Contact dealer technical support - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		has measured current flow above a specified range ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	
P215B-00	Vehicle Speed / Output Shaft Speed Correlation - Reserved by document	▪ Incorrect gearbox output shaft speed signal being sent from the transmission control module ▪ Incorrect vehicle speed signal being sent from anti-lock brake system ▪ Other anti-lock brake system control module related DTCs ▪ Other transmission related DTCs ▪ Driveline system mechanical integrity	▪ Check anti-lock brake system control module for related DTCs and refer to relevant DTC index ▪ Check transmission control module related DTCs and refer to relevant DTC index ▪ Check driveline system for mechanical integrity ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P215B-40	Vehicle Speed / Output Shaft Speed Correlation - No sub type information	▪ Incorrect low range signal being sent from the transmission control module ▪ Incorrect gearbox output shaft speed signal being sent from the transmission control	▪ Check anti-lock brake system control module for related DTCs and refer to relevant DTC index ▪ Check transmission control module related DTCs and refer to relevant DTC index ▪ Check driveline system for mechanical integrity ▪ Using the Jaguar Land Rover approved

		module ▪ Incorrect vehicle speed signal being sent from anti-lock brake system ▪ Other anti-lock brake system control module related DTCs ▪ Other transmission related DTCs ▪ Driveline system mechanical integrity	diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2183-2A	Engine Coolant Temperature Sensor 2 Circuit Range/Performance - Signal stuck in range	▪ Radiator outlet temperature sensor value has remained static within a plausible range ▪ Radiator outlet temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Radiator outlet temperature sensor failure	▪ Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data logs for signals ▪ 0x0488 Engine Coolant Temperature Sensor #2 Voltage Volts ▪ 0x0489 Engine Coolant Temperature #2 - Deg C ▪ Refer to the electrical circuit diagram and check the radiator outlet temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new radiator outlet temperature sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2183-84	Engine Coolant Temperature Sensor 2 Circuit Range/Performance - Signal below allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range	▪ Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data logs for signals ▪ 0x0488 Engine Coolant Temperature Sensor #2 Voltage Volts ▪ 0x0489 Engine Coolant

		▪ Radiator outlet temperature sensor circuit short circuit to ground, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Radiator outlet temperature sensor failure	Temperature #2 - Deg C ▪ Refer to the electrical circuit diagram and check the radiator outlet temperature sensor circuit for short circuit to ground, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new radiator outlet temperature sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P2183-85	Engine Coolant Temperature Sensor 2 Circuit Range/Performance - Signal above allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range ▪ Radiator outlet temperature sensor circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Radiator outlet temperature sensor failure	▪ Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data list signals ▪ 0x0488 Engine Coolant Temperature Sensor #2 Voltage Volts ▪ 0x0489 Engine Coolant Temperature #2 - Deg C ▪ Refer to the electrical circuit diagram and check the radiator outlet temperature sensor circuit for short circuit to power ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new radiator outlet temperature sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P2184-16	Engine Coolant Temperature Sensor 2 Circuit Low - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground	▪ Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data list signals ▪ 0x0488 Engine Coolant Temperature Sensor #2 Voltage Volts

		▪ Radiator outlet temperature sensor circuit short circuit to ground, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Radiator outlet temperature sensor failure	▪ 0x0489 Engine Coolant Temperature #2 - Deg C ▪ Refer to the electrical circuit diagram and check the radiator outlet temperature sensor circuit for short circuit to ground, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new radiator outlet temperature sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2185-17	Engine Coolant Temperature Sensor 2 Circuit High - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Radiator outlet temperature sensor circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Radiator outlet temperature sensor failure	▪ Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data list for signals ▪ 0x0488 Engine Coolant Temperature Sensor #2 Voltage Volts ▪ 0x0489 Engine Coolant Temperature #2 - Deg C ▪ Refer to the electrical circuit diagram and check the radiator outlet temperature sensor circuit for short circuit to power ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new radiator outlet temperature sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2195-00	O2 Sensor Signal Biased/Stuck Lean - Bank 1, Sensor 1 - No sub type information		▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagram and check the heated oxygen sensor heater circuit for short circuit to ground

		NOTE: Measured oxygen concentration implausibly high compared to model oxygen concentration for Sensor 1 Bank 1, during part load and during low part load - Other related DTCs - Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Heated oxygen sensor failure	short circuit to power, open circuit, resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new heated oxygen sensor as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Oxygen sensor' routine (0x4035) - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Diesel fuelling adaptation re-initialization' routine (0x4020) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P2195-85	O2 Sensor Signal Biased/Stuck Lean - Bank 1, Sensor 1 - Signal above allowable range	- The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range - Other related DTCs - Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open	- Check engine control module for fuel injection related DTCs and refer to relevant DTC index - Refer to the electrical circuit diagram and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new heated oxygen sensor as required - Using the Jaguar Land Rover

		circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Heated oxygen sensor failure	approved diagnostic equipment carry out 'Oxygen sensor' routine (0x4035) - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Diesel fuelling adaptation re-initialization' routine (0x4020) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take and retest
P2196-00	O2 Sensor Signal Biased/Stuck Rich - Bank 1, Sensor 1 - No sub type information	NOTE: Measured oxygen concentration implausibly low compared to model oxygen concentration for Sensor 1 Bank 1, during part load and during low part load - Other related DTCs - Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Heated oxygen sensor failure	- Check engine control module for fuel injection related DTCs and refer to relevant DTC index - Refer to the electrical circuit diagram and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new heated oxygen sensor as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Oxygen sensor' routine (0x4035) - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Diesel fuelling adaptation re-initialization' routine (0x4020) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take and retest
P2196-84	O2 Sensor Signal Biased/Stuck Rich - Bank 1,	Intake air distribution and	Check the intake air and distribution system for faults, air leaks and correct

	Sensor 1 - Signal below allowable range	filtering system component fault or air leaks ▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ▪ Other related DTCs ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	installation. Repair as required and retest ▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagram and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new heated oxygen sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out 'Oxygen sensor' routine (0x4035) ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out 'Diesel fuelling adaptation re-initialization' routine (0x4020) ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P21C5-64	Reductant Level Sensor "A" Stuck - Signal plausibility failure	▪ Purpose of the DTC ▪ To monitor DEF tank level sensor signal ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ DEF tank level sensor signal is not updated ▪ Prioritised List of Possible Causes ▪ The engine control module	▪ Vehicle Conditions to enable DTC Logging strategy ▪ DEF system pressurized ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		detected plausibility failures ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	
P21C7-13	Reductant Control Module Power Relay/Relays Control Circuit/Open - Circuit open	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid relay ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference SCRMRLY ▪ Monitor Description ▪ Diesel exhaust fluid relay short circuit open circuit, high resistance ▪ Prioritised List of Possible Causes ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The electrical checks normally occur once ignition is on or after engine start ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Diesel exhaust fluid relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid relay failure	▪ Check and install a new diesel exhaust fluid relay as required ▪ Contact dealer technical support ▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ On completion of the previous repair actions the following MU be carried out ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest ▪ NOTE: The engine MIL is OFF (when engine ON) ▪ Start engine ▪ The 'inducement' warning clear automatically once the monitor has executed and passed with no malfunction. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle, then the malfunction has re-occurred. Further repair is required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This clears the fault stored from P2BAE / P2BAF
P21C8-11	Reductant Control Module Power Relay/Relays Control Circuit Low - Circuit short to ground	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid relay ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference SCRMRLY ▪ Monitor Description	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid relay circuit for short circuit to ground, short circuit to

		▪ Diesel exhaust fluid relay circuit short circuit to ground ▪ Prioritised List of Possible Causes ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Diesel exhaust fluid relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid relay failure	power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid relay as required ▪ Contact dealer technical support ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P21C9-12	Reductant Control Module Power Relay/Relays Control Circuit High - Circuit short to battery	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid relay ▪ Electrical Cause	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit

			▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference SCRMRLY ▪ Monitor Description ▪ Diesel exhaust fluid relay circuit short circuit to power ▪ Prioritised List of Possible Causes ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Diesel exhaust fluid relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid relay as required ▪ Contact dealer technical support ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
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		▪ Diesel exhaust fluid relay failure	
P2200-00	NOx Sensor Circuit (Bank 1 Sensor 1) - No sub type information	▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	▪ Refer to the electrical circuit diagram and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2200-13	NOx Sensor Circuit (Bank 1 Sensor 1) - Circuit open	▪ Purpose of the DTC ▪ Circuit check of pre selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Pre selective catalytic reduction NOx sensor circuit open circuit, high resistance ▪ Prioritised List of Possible Causes ▪ The engine	▪ Vehicle Conditions to enable DTC Logging strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Pre selective catalytic reduction NOx sensor circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	
P2200-14	NOx Sensor Circuit (Bank 1 Sensor 1) - Circuit short to ground or open	▪ Purpose of the DTC ▪ Circuit check of pre selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Pre selective catalytic reduction NOx sensor short	▪ Vehicle Conditions to enable DTC Logging strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre

		circuit to ground ▪ Prioritised List of Possible Causes ▪ The engine control module has detected a ground measurement or open circuit for a period longer than expected or has detected a ground measurement or open circuit when another value was expected ▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	selective catalytic reduction NOx sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2201-00	NOx Sensor Circuit Range/Performance (Bank 1 Sensor 1) - No sub type information	▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin	▪ Refer to the electrical circuit diagram and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the Jaguar Land Rover approved

		- corrosion - Pre selective catalytic reduction NOx sensor failure	diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2201-28	NOx Sensor Circuit Range/Performance (Bank 1 Sensor 1) - Signal bias level out of range / zero adjustment failure	- Purpose of the DTC - Offset check of pre selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Pre selective catalytic reduction NOx sensor positive / negative offset detected during overrun - Prioritised List of Possible Causes - Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Pre selective catalytic	- Vehicle Conditions to enable DTC Logging strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new pre selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		reduction NOx sensor failure	
P2201-2F	NOx Sensor Circuit Range/Performance (Bank 1 Sensor 1) - Signal erratic	▪ Purpose of the DTC ▪ Feedback check of pre selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Pre selective catalytic reduction NOx sensor delivers unplausible signal under steady state ▪ Prioritised List of Possible Causes ▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	▪ Vehicle Conditions to enable DTC Logging strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

P2202-00	NOx Sensor Circuit Low (Bank 1 Sensor 1) - No sub type information	- Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Pre selective catalytic reduction NOx sensor failure	- Refer to the electrical circuit diagram and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new pre selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2202-16	NOx Sensor Circuit Low (Bank 1 Sensor 1) - Circuit voltage below threshold	- Purpose of the DTC - O2 range check of pre selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Pre selective catalytic reduction NOx sensor O2 signal not in range minimum - Prioritised List of Possible Causes - The engine control module measured a voltage below a	- Vehicle Conditions to enable DTC Logging strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new pre selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		specified range but not necessarily a short circuit to ground - Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Pre selective catalytic reduction NOx sensor failure	
P2203-17	NOx Sensor Circuit High (Bank 1 Sensor 1) - Circuit voltage above threshold	- Purpose of the DTC - O2 range check of pre selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Pre selective catalytic reduction NOx sensor O2 signal not in range maximum	- Vehicle Conditions to enable DTC Logging strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new pre selective catalytic reduction NOx sensor as required

		▪ Prioritised List of Possible Causes ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2209-01	NOx Sensor Heater Sense Circuit Range/Performance (Bank 1 Sensor 1) - General electrical failure	▪ Purpose of the DTC ▪ Heater performance check of pre selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H	▪ Vehicle Conditions to enable DTC Logging strategy ▪ NOx sensor heating on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment

		- Monitor Description - Pre selective catalytic reduction NOx sensor heater unavailable - Pre selective catalytic reduction NOx sensor heater is not able to reach temperature - Prioritised List of Possible Causes - Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Pre selective catalytic reduction NOx sensor failure	clear all stored DTCs using the Diagnosis Menu tab and retest
P220A-00	NOx Sensor Supply Voltage Circuit (Bank 1 Sensor 1) - No sub type information	- Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Refer to the electrical circuit diagram and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new pre selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab

		Pre selective catalytic reduction NOx sensor failure	and retest
P220A-01	NOx Sensor Supply Voltage Circuit (Bank 1 Sensor 1) - General electrical failure	- Purpose of the DTC - Circuit check of pre selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Pre selective catalytic reduction NOx sensor heater circuit error - Prioritised List of Possible Causes - Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Pre selective catalytic reduction NOx sensor failure	- Vehicle Conditions to enable DTC Logging strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new pre selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

P220B-00	NOx Sensor Supply Voltage Circuit (Bank 1 Sensor 2) - No sub type information	- Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Post selective catalytic reduction NOx sensor failure	- Refer to the electrical circuit diagram and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P220B-01	NOx Sensor Supply Voltage Circuit (Bank 1 Sensor 2) - General electrical failure	- Purpose of the DTC - Circuit check of post selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Post selective catalytic reduction NOx sensor heater circuit error - Prioritised List of Possible Causes - Post selective catalytic reduction NOx sensor circuit short circuit to ground, short	- Vehicle Conditions to enable DTC Logging strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Post selective catalytic reduction NOx sensor failure	
P2226-64	Barometric Pressure Sensor A Circuit - Signal plausibility failure	- The engine control module detected plausibility failures - Other related DTCs - Barometric pressure sensor is integral to the engine control module - Barometric pressure sensor 5 volt signal circuit, below minimum voltage expected - Barometric pressure sensor failure internal to the engine control module	- Check engine control module for related DTCs and refer to relevant index - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit - Check and install a new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'New Module - Power control module - Diesel' routine - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Fuel lift pump' routine (0x4027_00) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take test and retest
P2227-84	Barometric Pressure Sensor A Circuit Range/Performance - Signal below allowable range	- The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range - Other related DTCs - Barometric pressure sensor is integral to	- Check engine control module for related DTCs and refer to relevant index - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit - Check and install a new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment

		the engine control module - Barometric pressure sensor 5 volt signal circuit, below minimum voltage expected - Barometric pressure sensor failure internal to the engine control module	carry out 'New Module - Power control module - Diesel' routine - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Fuel lift pump' routine (0x4027_00) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P2227-85	Barometric Pressure Sensor A Circuit Range/Performance - Signal above allowable range	- The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range - Other related DTCs - Barometric pressure sensor is integral to the engine control module - Barometric pressure sensor 5 volt signal circuit, below minimum voltage expected - Barometric pressure sensor failure internal to the engine control module	- Check engine control module for related DTCs and refer to relevant index - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit - Check and install a new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'New Module - Power control module - Diesel' routine - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Fuel lift pump' routine (0x4027_00) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P2228-16	Barometric Pressure Sensor "A" Circuit Low - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Other related DTCs - Barometric pressure sensor is integral to	- Check engine control module for related DTCs and refer to relevant index - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit - Check and install a new engine control module as required - Using the Jaguar Land Rover

		the engine control module - Barometric pressure sensor 5 volt signal circuit, below minimum voltage expected - Barometric pressure sensor failure internal to the engine control module	approved diagnostic equipment carry out 'New Module - Power control module - Diesel' routine - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Fuel lift pump' routine (0x4027_00) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take and retest
P2229-17	Barometric Pressure Sensor "A" Circuit High - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Other related DTCs - Barometric pressure sensor is integral to the engine control module - Barometric pressure sensor 5 volt signal circuit, below minimum voltage expected - Barometric pressure sensor failure internal to the engine control module	- Check engine control module for related DTCs and refer to relevant index - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit - Check and install a new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'New Module - Power control module - Diesel' routine - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Fuel lift pump' routine (0x4027_00) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take and retest
P223C-00	O2 Sensor Pumping Current Range/Performance Bank 1 - No sub type information	- Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin	- Refer to the electrical circuit diagram and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new heated oxygen sensor as required - Using the Jaguar Land Rover

		corrosion Heated oxygen sensor failure	approved diagnostic equipment carry out 'Oxygen sensor' routine (0x4035) - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Diesel fuelling adaptation re-initialization' routine (0x4020) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take test and retest
P2251-13	O2 Sensor Negative Current Control Circuit / Open - Bank 1, Sensor 1 - Circuit open	- The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Heated oxygen sensor failure	- Refer to the electrical circuit diagram and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new heated oxygen sensor as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Oxygen sensor' routine (0x4035) - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Diesel fuelling adaptation re-initialization' routine (0x4020) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take test and retest
P225C-85	NOx Sensor Performance - Signal Biased/Stuck High Bank 1 Sensor 1 - Signal above allowable range	- Purpose of the DTC - Range check pre selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - No	- Vehicle Conditions to enable DTC Logging strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring)

		- Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Pre selective catalytic reduction NOx sensor signal out of range at low load positive deviation - Prioritised List of Possible Causes - The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range - Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Pre selective catalytic reduction NOx sensor failure	- Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new pre selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P22D5-36	Turbocharger Turbine Inlet Valve Position Sensor Circuit Range/Performance	The engine control module detected excessive duration	Diagnosis of this DTC may require the Jaguar Land Rover approved diagnostic equipment check data

		- Signal frequency too low	for one cycle of the output across a specified sample size - Variable geometry turbocharger vane actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Variable geometry turbocharger vane actuator failure	signals 0x0342 Boost Pressure Actuator Bank1 - Measured position - Refer to the electrical circuit diagram and check the variable geometry turbocharger vane actuator circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment perform route 1. Inline diagnostic unit 2 diagnostic test Turbocharger - Check and install a new variable geometry turbocharger vane actuator as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take test and retest
P22D5-37	Turbocharger Turbine Inlet Valve Position Sensor Circuit Range/Performance - Signal frequency too high	- The engine control module detected insufficient duration for one cycle of the output across a specified sample size - Variable geometry turbocharger vane actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Variable geometry turbocharger vane actuator failure	- Diagnosis of this DTC may require use of the Jaguar Land Rover approved diagnostic equipment check data list for signals 0x0342 Boost Pressure Actuator Bank1 - Measured position - Refer to the electrical circuit diagram and check the variable geometry turbocharger vane actuator circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment perform route 1. Inline diagnostic unit 2 diagnostic test Turbocharger - Check and install a new variable geometry turbocharger vane actuator as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take test and retest	

			and retest
P225D-84	NOx Sensor Performance - Signal Biased/Stuck Low Bank 1 Sensor 1 - Signal below allowable range	■ Purpose of the DTC ■ Range check pre selective catalytic reduction NOx sensor ■ Electrical Cause ■ Yes ■ Mechanical Cause ■ No ■ Control Module Cavity ■ Circuit reference CAN_L_3 ■ Circuit reference CAN_3_H ■ Monitor Description ■ Pre selective catalytic reduction NOx sensor signal out of range at low load negative deviation ■ Prioritised List of Possible Causes ■ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ■ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is	■ Vehicle Conditions to enable DTC Logging strategy ■ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ■ Prioritised Checks to Perform ■ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Check and install a new pre selective catalytic reduction NOx sensor as required ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		disconnected, connector pin is backed out, connector pin corrosion Pre selective catalytic reduction NOx sensor failure	
P225E-85	NOx Sensor Performance - Signal Biased/Stuck High Bank 1 Sensor 2 - Signal above allowable range	- Purpose of the DTC - Range check post selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Post selective catalytic reduction NOx sensor signal out of range at low load positive deviation - Prioritised List of Possible Causes - The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range - Post selective catalytic	- Vehicle Conditions to enable DTC Logging strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Post selective catalytic reduction NOx sensor failure	
P225F-84	NOx Sensor Performance - Signal Biased/Stuck Low Bank 1 Sensor 2 - Signal below allowable range	▪ Purpose of the DTC ▪ Range check post selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Post selective catalytic reduction NOx sensor signal out of range at low load negative deviation ▪ Prioritised List of Possible Causes ▪ The engine control module has determined failures where	▪ Vehicle Conditions to enable DTC Logging strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective catalytic reduction NOx sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		some circuit quantity, reported via serial data, is below a specified range - Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Post selective catalytic reduction NOx sensor failure	
P2265-64	Water in Fuel Sensor Circuit Range/Performance - Signal plausibility failure	- The engine control module detected plausibility failures - Water in fuel sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Water in fuel sensor failure	- Refer to the electrical circuit diagram and check the water in fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new water in fuel sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2269-68	Water in Fuel Condition - Event information	The engine control module indicated the detection of a system event that	- Drain water from water in fuel / filter sensor - Using the Jaguar Land Rover approved diagnostic equipment

		was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module Water in fuel sensor has detected water in fuel condition	carry out 'Reset water in fuel detection' routine (0x4064) Using the Jaguar Land Rover approach diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P226B-00	Turbocharger/Supercharger Boost Pressure Too High - Mechanical - No sub type information	- Error path indicating whether over boost shut down is active or not - This DTC is set when overboost shut down is active. This means that the vehicle has lost control of boost. The engine is shut down as a safety precaution. The engine control module checks for an overboost by system monitoring and also monitors vehicle acceleration, if in gear and engine acceleration at idle, also if in neutral - Other related DTCs - Fuel injection related DTCs - Exhaust system leakage - Excessive engine oil level - Excessive oil in the air induction system - Air induction system leakage - Charge air pressure sensor blocked	- Check engine control module for related DTCs and refer to relevant DTC index - Check engine control module for fuel injection related DTCs and refer to relevant DTC index - Check exhaust system for mechanical integrity - Check and correct the engine oil level - Check air induction system / turbocharger for failure relating to excessive charged air conditions - Check air induction system for leakage - Check air induction system for excessive oil - Check charge air pressure sensor for blockage - Check turbocharger system / vanes mechanical integrity - Using the Jaguar Land Rover approach diagnostic equipment perform routine Inline diagnostic unit 2 diagnostic test Turbocharger - Using the Jaguar Land Rover approach diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest

		▪ Turbocharger system / vanes mechanical integrity	
P226C-00	Turbocharger Boost Control "A" Slow Response - No sub type information	▪ Other related DTCs ▪ Fuel injection related DTCs ▪ Exhaust system leakage ▪ Excessive engine oil level ▪ Excessive oil in the air induction system ▪ Air induction system leakage ▪ Charge air pressure sensor blocked ▪ Turbocharger system / vanes mechanical integrity	▪ Check engine control module for related DTCs and refer to relevant index ▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index ▪ Check exhaust system for mechanical integrity ▪ Check and correct the engine oil level ▪ Check air induction system / turbocharger for failure relating to excessive charged air conditions ▪ Check air induction system for leakage ▪ Check air induction system for excessive oil ▪ Check charge air pressure sensor for blockage ▪ Check turbocharger system / vanes mechanical integrity ▪ Using the Jaguar Land Rover approved diagnostic equipment perform routine Inline diagnostic unit 2 diagnostic test Turbocharger ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P226D-00	Particulate Filter Deteriorated/Missing Substrate Bank 1 - No sub type information	▪ Other DPF related DTCs ▪ Diesel particulate substrate is missing - Severely damaged or removed ▪ DPF failure	NOTE: Operational requirements needed to allow the monitor to be fully tested. Drive at steady 70mph 6th gear for 10 minutes. Return to workshop and turn engine off, wait 5 minutes, key on and check for DTC ▪ Check engine control module for D related DTCs and refer to relevant index

			index - Remove the DPF and visually inspect for severe damage and correct installation - Check and install a new DPF as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Diesel particulate filter replacement' routine (0x405B_) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P2279-64	Intake Air System Leak - Signal plausibility failure	- The engine control module detected plausibility failures - Other related DTCs - Air induction system leakage - EGR system leakage - Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Heated oxygen sensor failure	- Check engine control module for related DTCs and refer to relevant index - Check air induction system for leakage - Check EGR system for leakage - Refer to the electrical circuit diagram and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new heated oxygen sensor as required - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Oxygen sensor' routine (0x4035) - Using the Jaguar Land Rover approved diagnostic equipment carry out 'Diesel fuelling adaptation re-initialization' routine (0x402C) - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P228E-21	Fuel Pressure Regulator 1 Exceeded Learning Limits - Too Low - Signal amplitude < minimum	The engine control module measured a signal voltage below a specified range but not necessarily a	Refer to the electrical circuit diagram and check the fuel pressure control valve circuit for short circuit to ground, short circuit to power, open circuit, resistance

		short circuit to ground, gain low ▪ Fuel pressure control valve circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel pressure control valve failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new fuel pressure control valve as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P228F-22	Fuel Pressure Regulator 1 Exceeded Learning Limits - Too High - Signal amplitude > maximum	▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high ▪ Fuel pressure control valve circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel pressure control valve failure	▪ Refer to the electrical circuit diagram and check the fuel pressure control valve circuit for short circuit to ground, short circuit to power, open circuit, resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new fuel pressure control valve as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P229E-00	NOx Sensor Circuit (Bank 1 Sensor 2) - No sub type information	▪ Post selective catalytic reduction NOx sensor circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out,	▪ Refer to the electrical circuit diagram and check the post selective catalytic reduction NOx sensor circuit for open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective

		- connector pin corrosion - Post selective catalytic reduction NOx sensor failure	catalytic reduction NOx sensor as required Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P229E-13	NOx Sensor Circuit (Bank 1 Sensor 2) - Circuit open	- Purpose of the DTC - Circuit check of post selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Post selective catalytic reduction NOx sensor circuit open circuit, high resistance - Prioritised List of Possible Causes - The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Post selective catalytic reduction NOx sensor circuit open circuit,	- Vehicle Conditions to enable DTC Logging strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Post selective catalytic reduction NOx sensor failure	
P229E-14	NOx Sensor Circuit (Bank 1 Sensor 2) - Circuit short to ground or open	- Purpose of the DTC - Circuit check of post selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Post selective catalytic reduction NOx sensor short circuit to ground - Prioritised List of Possible Causes - The engine control module has detected a ground measurement or open circuit for a period longer than expected or has detected a ground measurement or open circuit	- Vehicle Conditions to enable DTC Logging strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		when another value was expected - Post selective catalytic reduction NOx sensor circuit short circuit to ground, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Post selective catalytic reduction NOx sensor failure	
P229F-00	NOx Sensor Circuit Range/Performance (Bank 1 Sensor 2) - No sub type information	- Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Post selective catalytic reduction NOx sensor failure	- Refer to the electrical circuit diagram and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu take test and retest
P229F-28	NOx Sensor Circuit Range/Performance (Bank 1 Sensor 2) - Signal bias level out of range / zero adjustment failure	- Purpose of the DTC - Offset check of post selective catalytic reduction NOx sensor - Electrical Cause - Yes	- Vehicle Conditions to enable DTC Logging strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and

		▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Post selective catalytic reduction NOx sensor positive / negative offset detected during overrun ▪ Prioritised List of Possible Causes ▪ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Post selective catalytic reduction NOx sensor failure	then releases it, then the sensor heated and starts measuring) ▪ Overrun ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective catalytic reduction NOx sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P229F-2F	NOx Sensor Circuit Range/Performance (Bank 1 Sensor 2) - Signal erratic	▪ Purpose of the DTC ▪ Feedback check of post selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No	▪ Vehicle Conditions to enable DTC Logging strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor heated and starts measuring)

		- Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Post selective catalytic reduction NOx sensor delivers unplausible signal under steady state - Prioritised List of Possible Causes - Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Post selective catalytic reduction NOx sensor failure	- Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P22A0-00	NOx Sensor Circuit Low (Bank 1 Sensor 2) - No sub type information	- Post selective catalytic reduction NOx sensor circuit short circuit to ground, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Post selective	- Refer to the electrical circuit diagram and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved

		catalytic reduction NOx sensor failure	diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P22A0-16	NOx Sensor Circuit Low (Bank 1 Sensor 2) - Circuit voltage below threshold	■ Purpose of the DTC ■ O2 range check of post selective catalytic reduction NOx sensor ■ Electrical Cause ■ Yes ■ Mechanical Cause ■ No ■ Control Module Cavity ■ Circuit reference CAN_L_3 ■ Circuit reference CAN_3_H ■ Monitor Description ■ Post selective catalytic reduction NOx sensor O2 signal not in range minimum ■ Prioritised List of Possible Causes ■ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ■ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is	■ Vehicle Conditions to enable DTC Logging strategy ■ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ■ Prioritised Checks to Perform ■ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Check and install a new post selective catalytic reduction NOx sensor as required ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		disconnected, connector pin is backed out, connector pin corrosion Post selective catalytic reduction NOx sensor failure	
P22A1-17	NOx Sensor Circuit High (Bank 1 Sensor 2) - Circuit voltage above threshold	- Purpose of the DTC - O2 range check of post selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Post selective catalytic reduction NOx sensor O2 signal not in range maximum - Prioritised List of Possible Causes - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Post selective catalytic reduction NOx sensor circuit	- Vehicle Conditions to enable DTC Logging strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Post selective catalytic reduction NOx sensor failure	
P22A7-01	NOx Sensor Heater Sense Circuit Range/Performance (Bank 1 Sensor 2) - General electrical failure	▪ Purpose of the DTC ▪ Heater performance check of post selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Post selective catalytic reduction NOx sensor heater unavailable ▪ Post selective catalytic reduction NOx sensor heater is not able to reach temperature ▪ Prioritised List of	▪ Vehicle Conditions to enable DTC Logging strategy ▪ NOx sensor heating on ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective catalytic reduction NOx sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		Possible Causes ▪ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Post selective catalytic reduction NOx sensor failure	
P22C5-13	Turbocharger Compressor Outlet Valve Control Circuit / Open - Circuit open	▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Charge air shut-off valve circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Charge air shut-off valve failure	▪ Refer to the electrical circuit diagram and check the charge air shut-off valve circuit for open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new charge air shut-off valve as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P22C6-11	Turbocharger Compressor Outlet Valve Control Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has	▪ Refer to the electrical circuit diagram and check the charge air shut-off valve circuit for short circuit to ground ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		detected a ground measurement when another value was expected - Charge air shut-off valve circuit short circuit to ground - Connector is disconnected, connector pin is backed out, connector pin corrosion - Charge air shut-off valve failure	corrosion - Check and install a new charge air shut-off valve as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P22C7-12	Turbocharger Compressor Outlet Valve Control Circuit High - Circuit short to battery	- The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Charge air shut-off valve circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - Charge air shut-off valve failure	- Refer to the electrical circuit diagram and check the charge air shut-off valve circuit for short circuit to power - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new charge air shut-off valve as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P22FA-27	NOx Sensor Performance - Slow Response High to Low Bank 1 Sensor 1 - Signal rate of change above threshold	- Purpose of the DTC - Dynamic check of pre selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause	- Vehicle Conditions to enable DTC Logging strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor

		▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Pre selective catalytic reduction NOx sensor slow response to dynamic transients ▪ Prioritised List of Possible Causes ▪ The signal transitions more quickly than is reasonably allowed ▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P22FB-64	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1 - Signal plausibility failure	▪ Purpose of the DTC ▪ Plausibility check of pre selective catalytic reduction NOx sensor	▪ Vehicle Conditions to enable DTC Logging strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the

		▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Pre selective catalytic reduction NOx sensor positive / negative deviation to the modelled value ▪ Prioritised List of Possible Causes ▪ The engine control module detected plausibility failures ▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	function integrates the warmth quantity seen by the sensor and then releases it, then the sensor heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
P22FB-84	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1 - Signal Below	▪ Purpose of the DTC ▪ Out of range check of pre	▪ Vehicle Conditions to enable DTC Logging strategy ▪ NOx sensor released, the dew

Allowable Range

selective catalytic reduction NOx sensor

- Electrical Cause
 - Yes
- Mechanical Cause
 - No
- Control Module Cavity
 - Circuit reference CAN_L_3
 - Circuit reference CAN_3_H
- Monitor Description
 - Pre selective catalytic reduction NOx sensor signal out of range at low load negative deviation
 - Pre selective catalytic reduction NOx sensor signal range check is less than 100ppm
- Prioritised List of Possible Causes
 - The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range
 - Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit,

point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring)

- Prioritised Checks to Perform
 - Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
 - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
 - Check and install a new pre selective catalytic reduction NOx sensor as required
 - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Pre selective catalytic reduction NOx sensor failure	
P22FB-85	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1 - Signal above allowable range	- Purpose of the DTC - Out of range check of pre selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Pre selective catalytic reduction NOx sensor signal out of range at low load positive deviation - Pre selective catalytic reduction NOx sensor signal range check is larger than 3000ppm - Prioritised List of Possible Causes - The engine control module	- Vehicle Conditions to enable DTC Logging strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new pre selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		has determined failures where some circuit quantity, reported via serial data, is above a specified range ▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	
P22FE-00	NOx Sensor Performance - Sensing Element Bank 1 Sensor 2 - No sub type information	▪ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Post selective catalytic reduction NOx sensor failure	▪ Refer to the electrical circuit diagram and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective catalytic reduction NOx sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu task and retest
P22FE-2A	NOx Sensor Performance - Sensing Element Bank 1 Sensor 2 - Signal stuck in	▪ Purpose of the DTC ▪ Post selective catalytic	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Engine running. NOx peak

	range	reduction NOx sensor plausibility ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Post selective catalytic reduction NOx sensor signal is not plausible in part load ▪ Prioritised List of Possible Causes ▪ Post selective catalytic reduction NOx sensor no reaction on high NOx peak ▪ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Post selective catalytic reduction NOx sensor failure	▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective catalytic reduction NOx sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
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P22FE-64	NOx Sensor Performance - Sensing Element Bank 1 Sensor 2 - Signal plausibility failure	■ Purpose of the DTC ■ Post selective catalytic reduction NOx sensor removal detection ■ Electrical Cause ■ Yes ■ Mechanical Cause ■ Yes ■ Control Module Cavity ■ Circuit reference CAN_L_3 ■ Circuit reference CAN_3_H ■ Monitor Description ■ Post selective catalytic reduction NOx sensor signal is not plausible in part load ■ Prioritised List of Possible Causes ■ The engine control module detected plausibility failures ■ Post selective catalytic reduction NOx sensor not installed into exhaust system ■ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out,	■ Vehicle Conditions to enable DTC Logging strategy ■ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ■ Overrun ■ Part load ■ Prioritised Checks to Perform ■ Check the post selective catalytic reduction NOx sensor is correctly installed into exhaust system ■ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Check and install a new post selective catalytic reduction NOx sensor as required ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest
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		connector pin corrosion Post selective catalytic reduction NOx sensor failure	
P22FE-84	NOx Sensor Performance - Sensing Element Bank 1 Sensor 2 - Signal below allowable range	- Purpose of the DTC - Out of range check of post selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Post selective catalytic reduction NOx sensor signal out of range at low load negative deviation - Post selective catalytic reduction NOx sensor signal range check is less than 100ppm - Prioritised List of Possible Causes - The engine control module has determined failures where some circuit quantity, reported via	- Vehicle Conditions to enable DTC Logging strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the Diagnosis Menu tab and retest

		serial data, is below a specified range - Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Post selective catalytic reduction NOx sensor failure	
P22FE-85	NOx Sensor Performance - Sensing Element Bank 1 Sensor 2 - Signal above allowable range	- Purpose of the DTC - Out of range check of post selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Post selective catalytic reduction NOx sensor signal out of range at low load positive deviation	- Vehicle Conditions to enable DTC Logging strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the Jaguar Land Rover approved diagnostic equipment

		■ Post selective catalytic reduction NOx sensor signal range check is larger than 3000ppm ■ Prioritised List of Possible Causes ■ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range ■ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Post selective catalytic reduction NOx sensor failure	clear all stored DTCs using the Diagnosis Menu tab and retest
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